

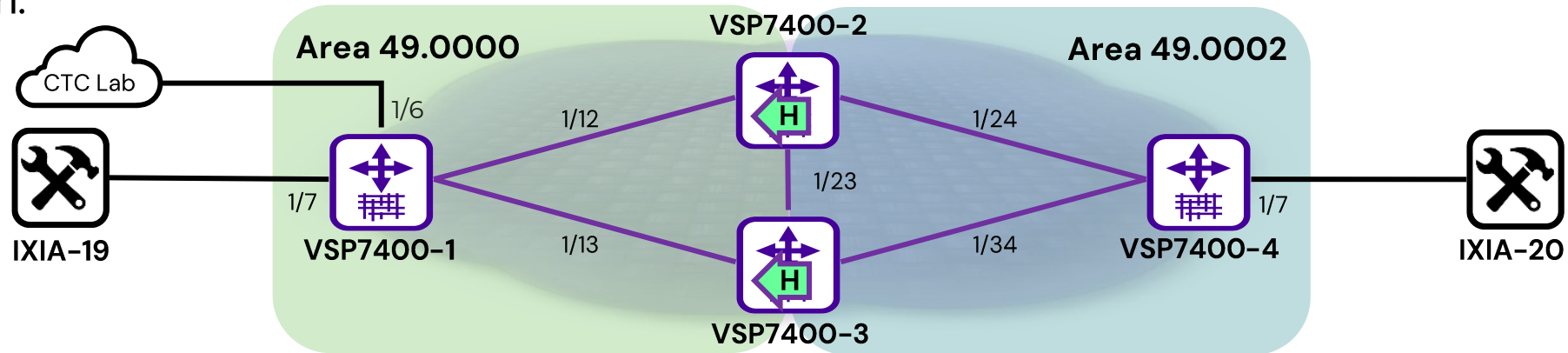
Multi-area Boundary Nodes Flipping home & remote areas

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Solutions Engineering
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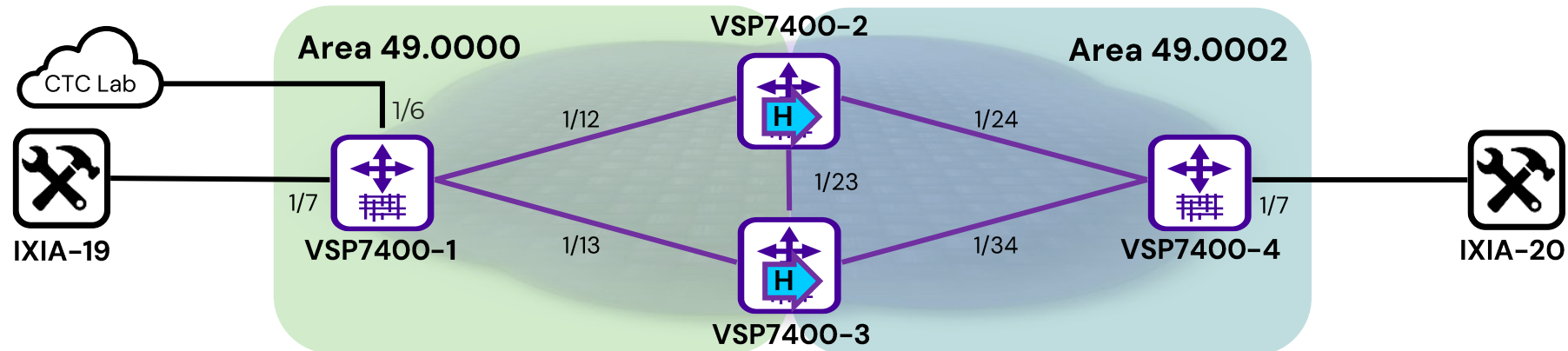
- It is normally best practice for Multi-area boundary nodes to have the HOME area facing the access area, rather than a backbone area
- This is because all services present on the boundary nodes are natively present in the HOME area, including Nickname server
- Also, when using auto-sense uplinks from fabric access edge nodes, the boundary nodes are only capable to handle auto-sense NNI into the HOME area; this becomes even more important when ISIS Hello authentication keys are in use, because it requires the boundary nodes to handle onboarding of defaulted fabric edge switches on auto-sense ports
- But, when carving out new ISIS areas in existing single area designs it is often easier to dynamically create a new REMOTE area on nodes which already have a HOME area, rather than the reverse: re-configuring the existing HOME area as REMOTE area and then creating the new ISIS area as HOME
- These slides investigate the feasibility of flipping the HOME and REMOTE areas on already configured multi-area boundary nodes

Goal

- How to flip ISIS Areas on already deployed Multi-area boundary nodes
- Inception:

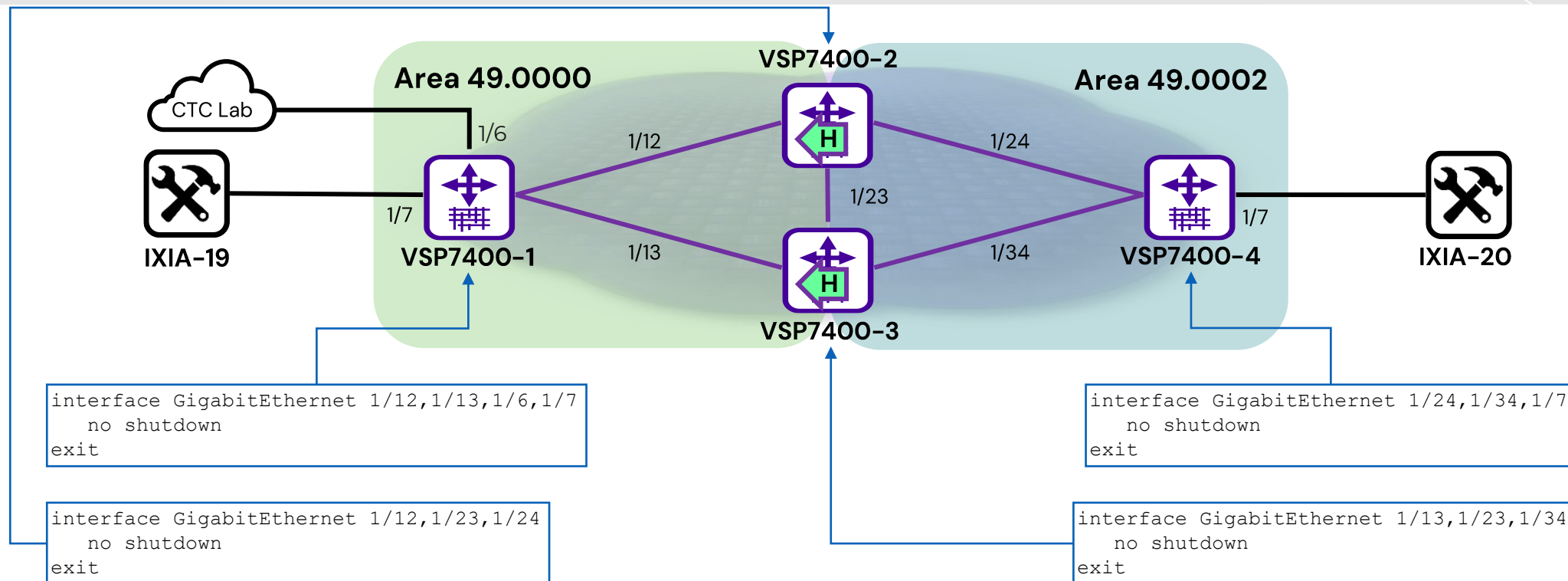


- Target:

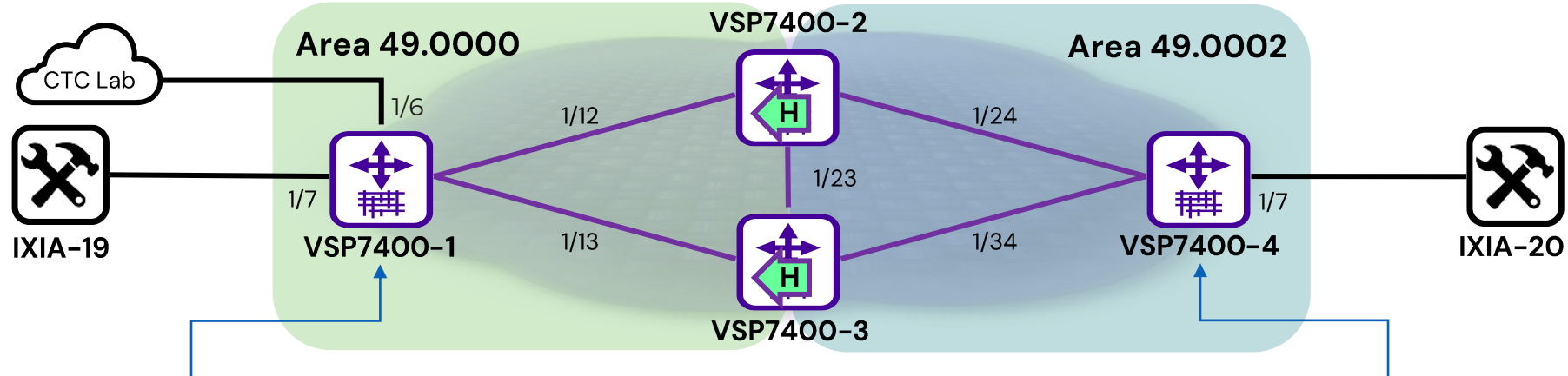


- All switches running 9.4.0.0GA for these tests

Inception: interfaces used



Inception: fabric config



```
spbm
router isis
  spbm 1
  spbm 1 nick-name 1.00.71
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7100
  manual-area 49.0000
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
cfm spbm enable
interface GigabitEthernet 1/12,1/13
  isis
  isis spbm 1
  isis hello-auth type hmac-md5 key fabricisspb
  isis enable
exit
```

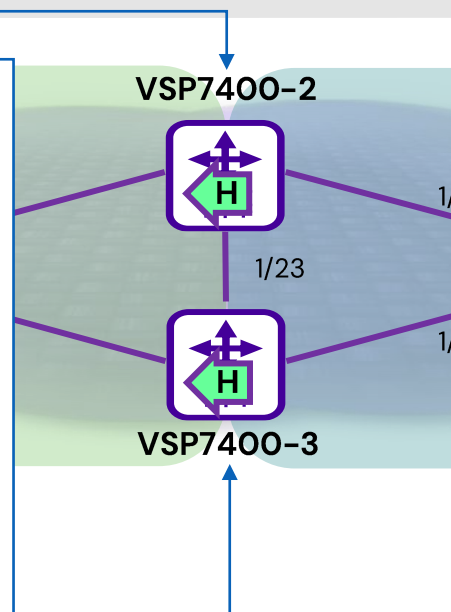
```
spbm
router isis
  spbm 1
  spbm 1 nick-name 2.00.74
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7400
  manual-area 49.0002
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
cfm spbm enable
interface GigabitEthernet 1/24,1/34
  isis
  isis spbm 1
  isis hello-auth type hmac-md5 key fabricisspb
  isis enable
exit
```

Inception: multi-area fabric config

```

spbm
router isis
  spbm 1
  spbm 1 nick-name 1.00.72
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7200
  manual-area 49.0000
  area-name green-area
  area-vnode sys-name Vnode-49.0000
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
router isis remote
  spbm 1 nick-name 2.00.72
  system-id e2bb.0000.7200
  manual-area 49.0002
  area-name blue-area
  area-vnode sys-name Vnode-49.0002
exit
router isis remote enable
cfm spbm enable
interface GigabitEthernet 1/12,1/23
  isis
  isis spbm 1
  isis hello-auth type hmac-md5 key fabricisspb
  isis enable
exit
interface GigabitEthernet 1/23,1/24
  isis remote
  isis remote spbm 1
  isis remote hello-auth type hmac-md5 key fabricisspb
  isis remote enable
exit

```

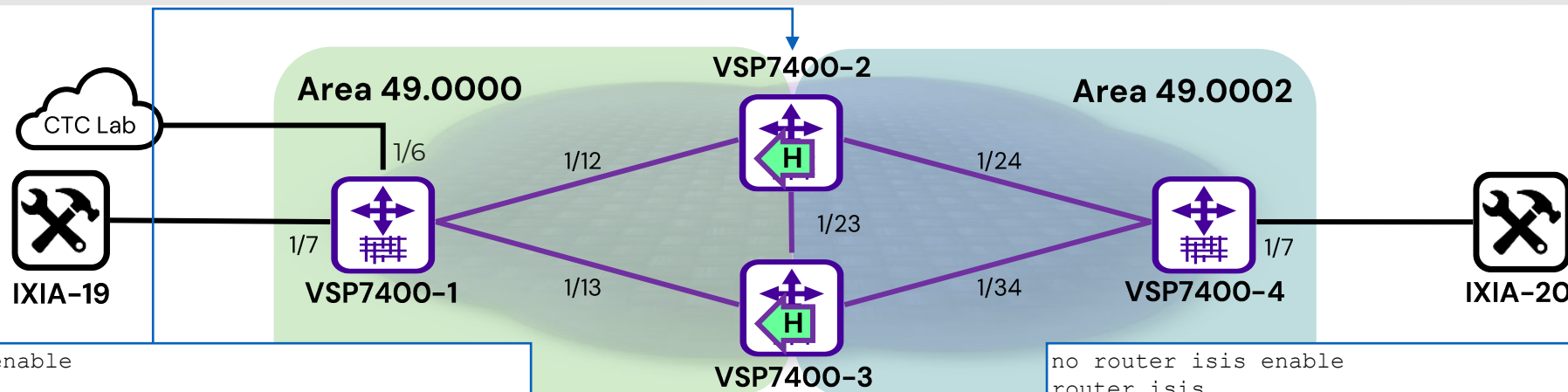


```

spbm
router isis
  spbm 1
  spbm 1 nick-name 1.00.73
  spbm 1 b-vid 4051-4052 primary 4051
  system-id 02bb.0000.7300
  manual-area 49.0000
  area-name green-area
  area-vnode sys-name Vnode-49.0000
exit
vlan create 4051 name "B-VLAN-1" type spbm-bvlan
vlan create 4052 name "B-VLAN-2" type spbm-bvlan
router isis enable
router isis remote
  spbm 1 nick-name 2.00.73
  system-id e2bb.0000.7300
  manual-area 49.0002
  area-name blue-area
  area-vnode sys-name Vnode-49.0002
exit
router isis remote enable
cfm spbm enable
interface GigabitEthernet 1/23,1/13
  isis
  isis spbm 1
  isis hello-auth type hmac-md5 key fabricisspb
  isis enable
exit
interface GigabitEthernet 1/23,1/34
  isis remote
  isis remote spbm 1
  isis remote hello-auth type hmac-md5 key fabricisspb
  isis remote enable
exit

```

Inception: Virtual-IST config (optional)

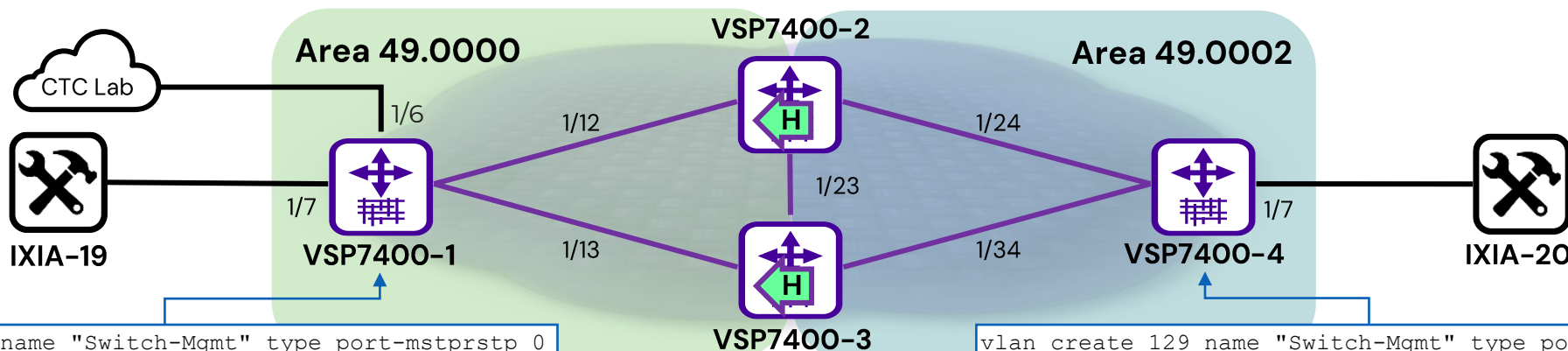


```
no router isis enable
router isis
  spbm 1 smlt-peer-system-id 02bb.0000.7300
  spbm 1 smlt-virtual-bmac 02:bb:00:00:72:ff
exit
vlan create 4000 name "IST-VLAN" type port-mstprstp 0
vlan i-sid 4000 15072073
interface Vlan 4000
  ip address 192.168.255.1/30
exit
virtual-ist peer-ip 192.168.255.2 vlan 4000
router isis enable
ip prefix-list "IST" 192.168.255.0/30
route-map "Suppress-IST" 1
  no permit
  enable
  match network "IST"
exit
route-map "Suppress-IST" 2
  permit
  enable
exit
```

- A Virtual-IST may or may not exist on the multi-area boundary nodes
- But if it does exist, the vIST connection runs in the HOME area, so the Smlt-Virt-BMAC (and Smlt-Peer-BMAC) will need to change when we flip the areas

```
no router isis enable
router isis
  spbm 1 smlt-peer-system-id 02bb.0000.7200
  spbm 1 smlt-virtual-bmac 02:bb:00:00:72:ff
exit
vlan create 4000 name "IST-VLAN" type port-mstprstp 0
vlan i-sid 4000 15072073
interface Vlan 4000
  ip address 192.168.255.2/30
exit
virtual-ist peer-ip 192.168.255.1 vlan 4000
router isis enable
ip prefix-list "IST" 192.168.255.0/30
route-map "Suppress-IST" 1
  no permit
  enable
  match network "IST"
exit
route-map "Suppress-IST" 2
  permit
  enable
exit
```

Inception: switch mgmt config

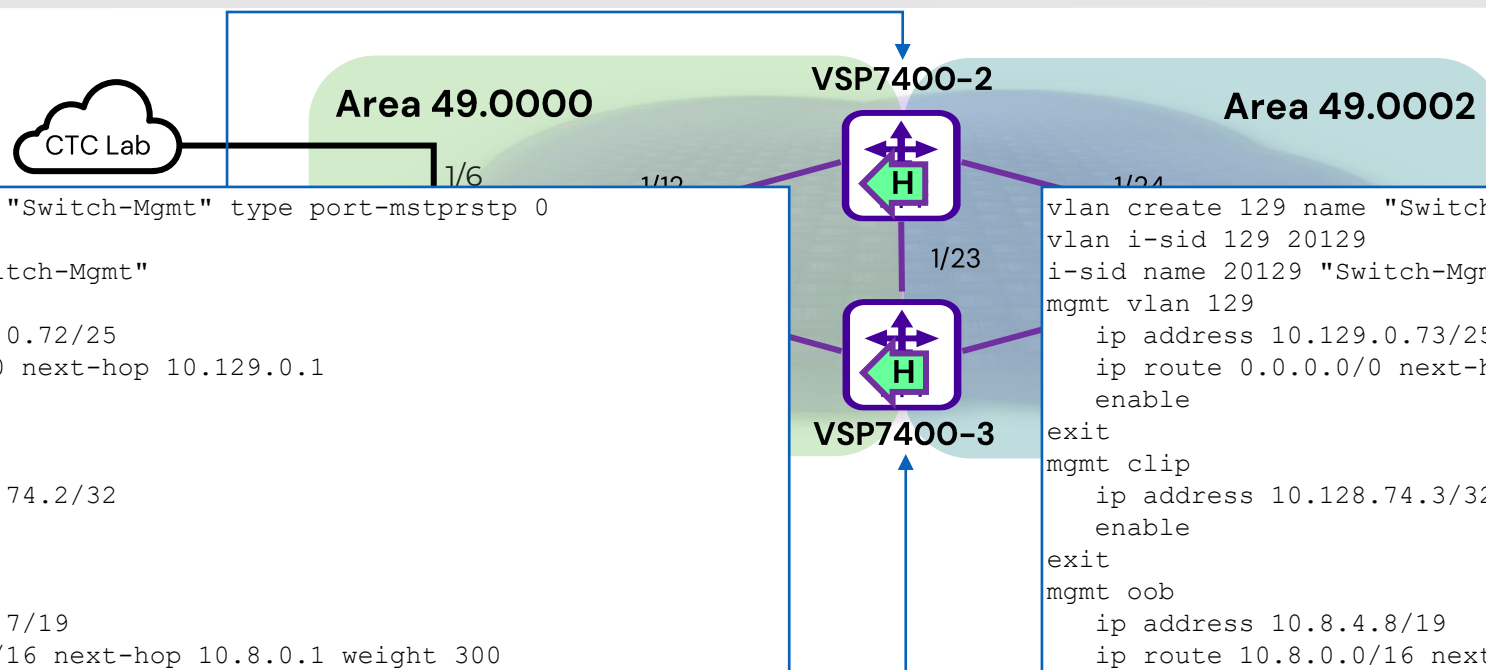


```
vlan create 129 name "Switch-Mgmt" type port-mstprstp 0
vlan i-sid 129 20129
i-sid name 20129 "Switch-Mgmt"
interface Vlan 129
  ip address 10.129.0.1 255.255.255.128
  mgmt enable
exit
mgmt clip
  ip address 10.128.74.1/32
  enable
exit
mgmt oob
  ip address 10.8.4.6/19
  ip route 10.8.0.0/16 next-hop 10.8.0.1 weight 300
  enable
exit
router isis
  spbm 1 ip enable
  redistribute direct
  redistribute direct enable
exit
isis apply redistribute direct
```

- Switches setup with a mgmt IP on each of: VLAN, CLIP, OOB
- This is to verify connectivity for inband mgmt (VLAN,CLIP) when flipping the areas

```
vlan create 129 name "Switch-Mgmt" type port-mstprstp 0
vlan i-sid 129 20129
i-sid name 20129 "Switch-Mgmt"
mgmt vlan 129
  ip address 10.129.0.74/25
  ip route 0.0.0.0/0 next-hop 10.129.0.1
  enable
exit
mgmt clip
  ip address 10.128.74.4/32
  enable
exit
mgmt oob
  ip address 10.8.4.9/19
  ip route 10.8.0.0/16 next-hop 10.8.0.1 weight 300
  enable
exit
router isis
  spbm 1 ip enable
  redistribute direct
  redistribute direct enable
exit
isis apply redistribute direct
```


Inception: switch mgmt config

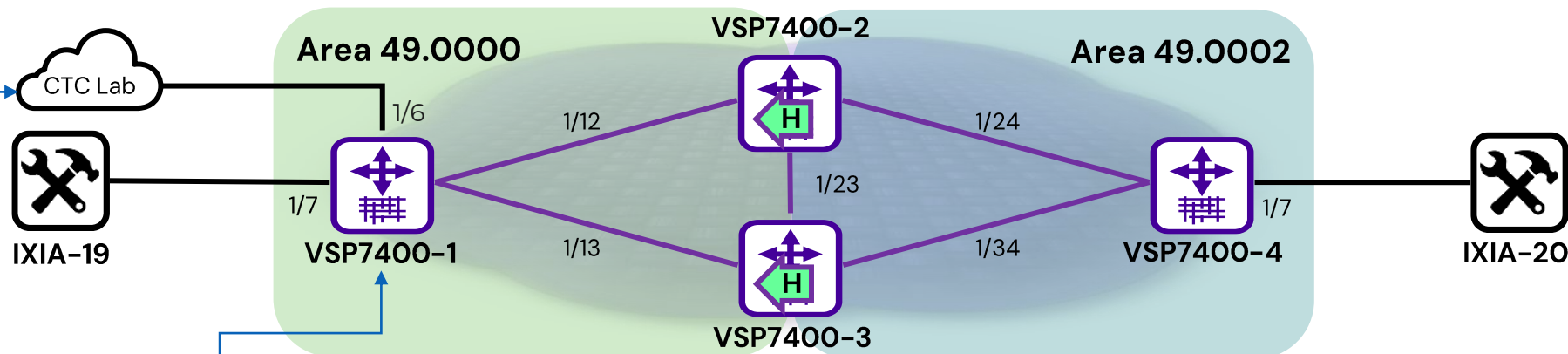


The multi-area redistribution is necessary for VSP7400-4 to be reachable from area 49.0000 for mgmt CLIP and VLAN interfaces. But will also be necessary for boundary node reachability for mgmt CLIP and VLAN IPs once we flip the areas and 49.0002 becomes the Home area

```
vlan create 129 name "Switch-Mgmt" type port-mstprstp 0
vlan i-sid 129 20129
i-sid name 20129 "Switch-Mgmt"
mgmt vlan 129
  ip address 10.129.0.72/25
  ip route 0.0.0.0/0 next-hop 10.129.0.1
  enable
exit
mgmt clip
  ip address 10.128.74.2/32
  enable
exit
mgmt oob
  ip address 10.8.4.7/19
  ip route 10.8.0.0/16 next-hop 10.8.0.1 weight 300
  enable
exit
router isis
  spbm 1 ip enable
  redistribute direct
  redistribute direct route-map "Suppress-IST"
  redistribute direct enable
  multi-area 12 isid-list allowed 20129
  multi-area 12 redistribute i-sid deny-all except-isid-list allowed
  multi-area ip redistribute unicast
  multi-area ip redistribute unicast enable
exit
isis apply redistribute direct
isis multi-area 12 apply redistribute i-sid
isis multi-area ip apply redistribute unicast
```

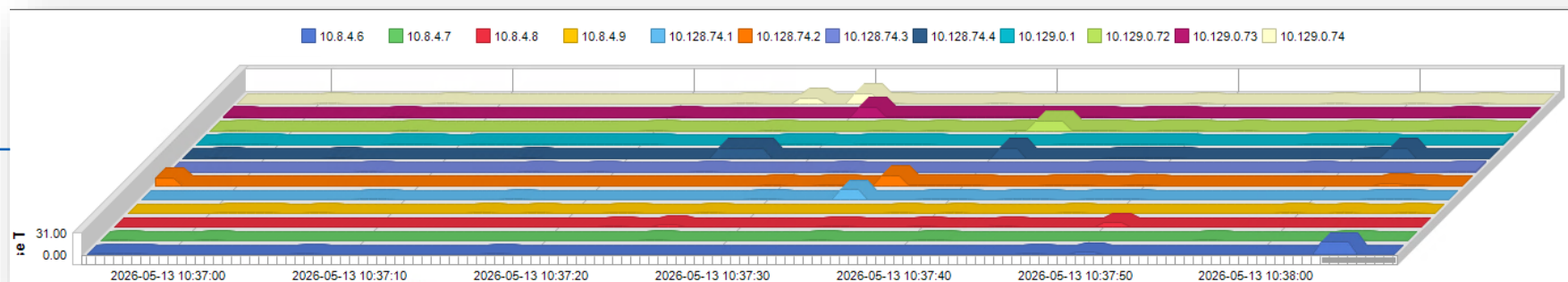
```
vlan create 129 name "Switch-Mgmt" type port-mstprstp 0
vlan i-sid 129 20129
i-sid name 20129 "Switch-Mgmt"
mgmt vlan 129
  ip address 10.129.0.73/25
  ip route 0.0.0.0/0 next-hop 10.129.0.1
  enable
exit
mgmt clip
  ip address 10.128.74.3/32
  enable
exit
mgmt oob
  ip address 10.8.4.8/19
  ip route 10.8.0.0/16 next-hop 10.8.0.1 weight 300
  enable
exit
router isis
  spbm 1 ip enable
  redistribute direct
  redistribute direct route-map "Suppress-IST"
  redistribute direct enable
  multi-area 12 isid-list allowed 20129
  multi-area 12 redistribute i-sid deny-all except-isid-list allowed
  multi-area ip redistribute unicast
  multi-area ip redistribute unicast enable
exit
isis apply redistribute direct
isis multi-area 12 apply redistribute i-sid
isis multi-area ip apply redistribute unicast
```

Inception: VSP7400-1 as IP router into wider CTC lab

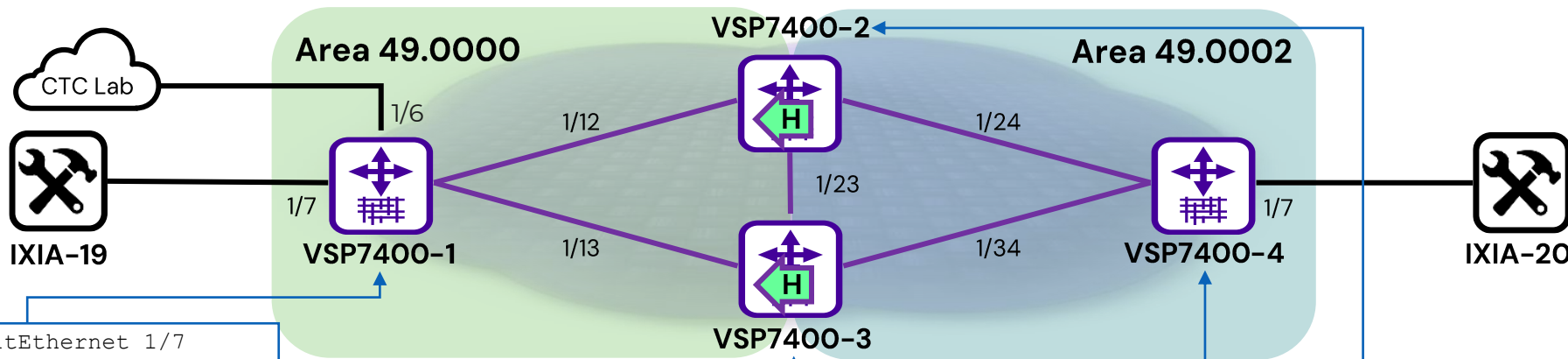


```
interface GigabitEthernet 1/6
  brouter port 1/6 vlan 3999 subnet 10.8.4.130/255.255.224.0
exit
ip route 0.0.0.0 0.0.0.0 10.8.0.1 weight 10
router isis
  spbm 1 ip enable
  redistribute static
  redistribute static enable
exit
isis apply redistribute static
```

- In band connectivity (mgmt VLAN & mgmt CLIP) to all switches in the setup will be via the CTC connection on VSP7400-1
- From CTC, all mgmt IPs (OOB,CLIP,VLAN) are reachable



Inception: VSP7400-1 as IP router into wider CTC lab



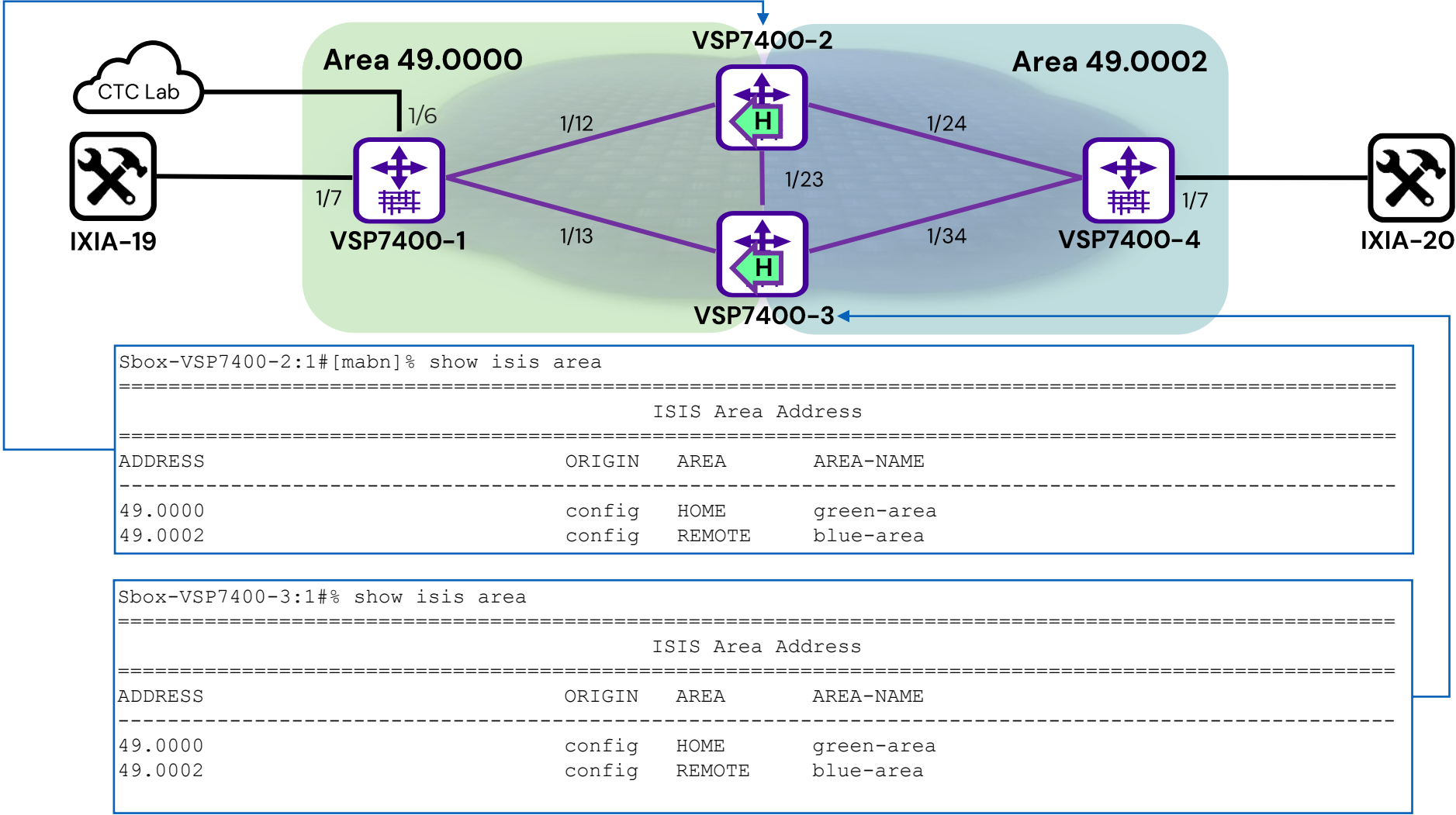
```
interface GigabitEthernet 1/7
 encapsulation dot1q
exit
vlan create 10 type port-mstprstp 0
vlan members add 10 1/7
vlan i-sid 10 20010
vlan create 11 type port-mstprstp 0
vlan members add 11 1/7
vlan i-sid 11 20011
```

```
router isis
 multi-area 12 isid-list allowed 20010,20011
 multi-area 12 redistribute i-sid deny-all except-isid-list allowed
exit
isis multi-area 12 apply redistribute i-sid
```

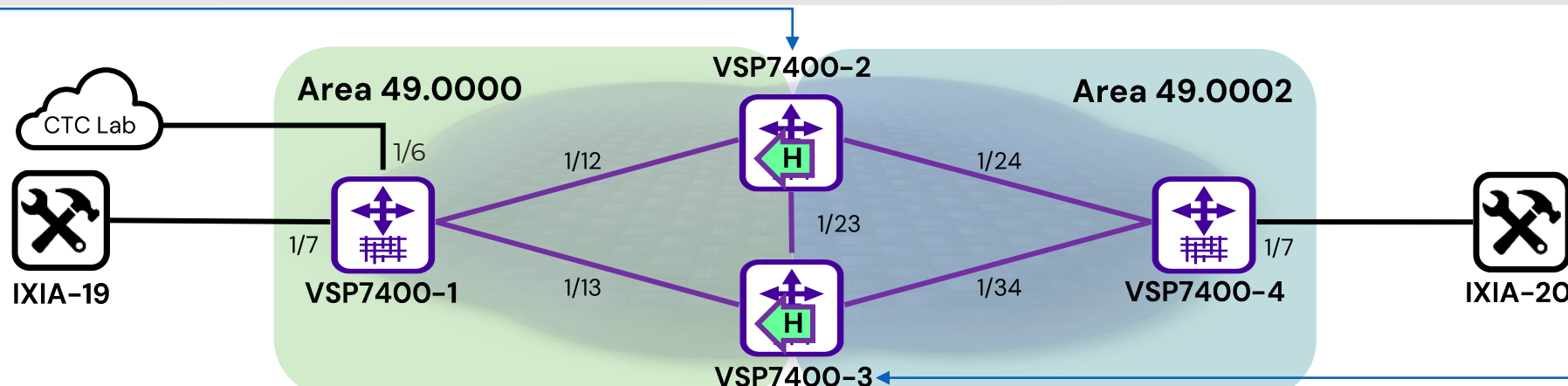
PGID	Total # Frames Rec.	Latency Avg (µs)	Byte Count	Frame Rate (/sec)	In Order Packets	Reordered Packets	Duplicate Packets	Late Packets	Lost Packets
Chassis: 10.8.8.2, Card: 0, Port: 019									
1	291,756	10.173	29,175,600	1,000	291,756	0	0	0	0
Chassis: 10.8.8.2, Card: 01, Port: 020									
0	291,757	10.144	29,175,700	1,000	291,757	0	0	0	0

- A couple of L2VSN services are created, and the IXIAs transmit a bidirectional regular 1000ps flow of traffic across both L2VSNs combined
 - 1 packet loss = 1 millisecond interruption
- This will be used to measure traffic interruptions while we flip the boundary node home/remote areas
- x2 L2VSNs, with even & odd I-SIDs as each will be hashed over one of the two multiarea boundary nodes

Inception: status



Inception: ISIS adjacencies



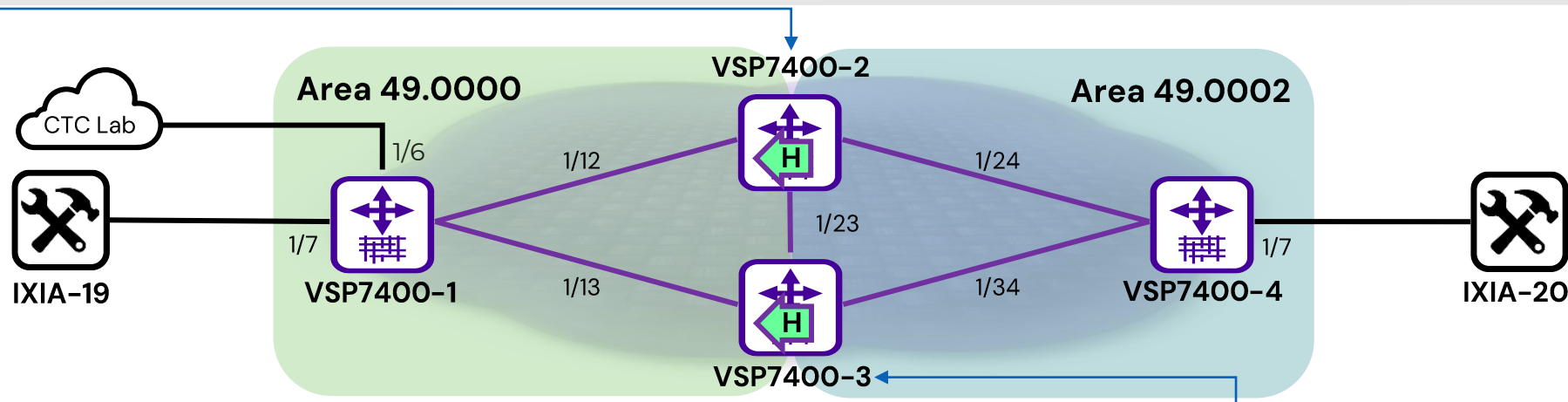
```
Sbox-VSP7400-2:1#[mabn]% show isis adjacencies
```

ISIS Adjacencies										
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA	AREA-NAME
Port1/12	1	UP	00:03:51	127		26 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	HOME	green-area
Port1/23	1	UP	00:03:51	127		24 02bb.0000.7300	Sbox-VSP7400-3	ACTIVE	HOME	green-area
PortVirtual	1	UP	00:03:39	127		27 9200.02ff.fff0	Vnode-49.0002	ACTIVE	HOME	green-area
Port1/23	1	UP	00:03:51	127		21 e2bb.0000.7300	Sbox-VSP7400-3	ACTIVE	REMOTE	blue-area
Port1/24	1	UP	00:03:51	127		21 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	REMOTE	blue-area
PortVirtual	1	UP	00:03:39	127		27 9200.00ff.fff0	Vnode-49.0000	ACTIVE	REMOTE	blue-area

```
Sbox-VSP7400-3:1#% show isis adjacencies
```

ISIS Adjacencies										
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA	AREA-NAME
Port1/13	1	UP	00:03:51	127		21 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	HOME	green-area
Port1/23	1	UP	00:03:51	127		21 02bb.0000.7200	Sbox-VSP7400-2	ACTIVE	HOME	green-area
PortVirtual	1	UP	00:03:39	127		27 9200.02ff.fff0	Vnode-49.0002	ACTIVE	HOME	green-area
Port1/23	1	UP	00:03:51	127		27 e2bb.0000.7200	Sbox-VSP7400-2	ACTIVE	REMOTE	blue-area
Port1/34	1	UP	00:03:51	127		26 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	REMOTE	blue-area
PortVirtual	1	UP	00:03:39	127		27 9200.00ff.fff0	Vnode-49.0000	ACTIVE	REMOTE	blue-area

Inception: status



- The designated boundary node is always the node with smallest chassis Base MAC

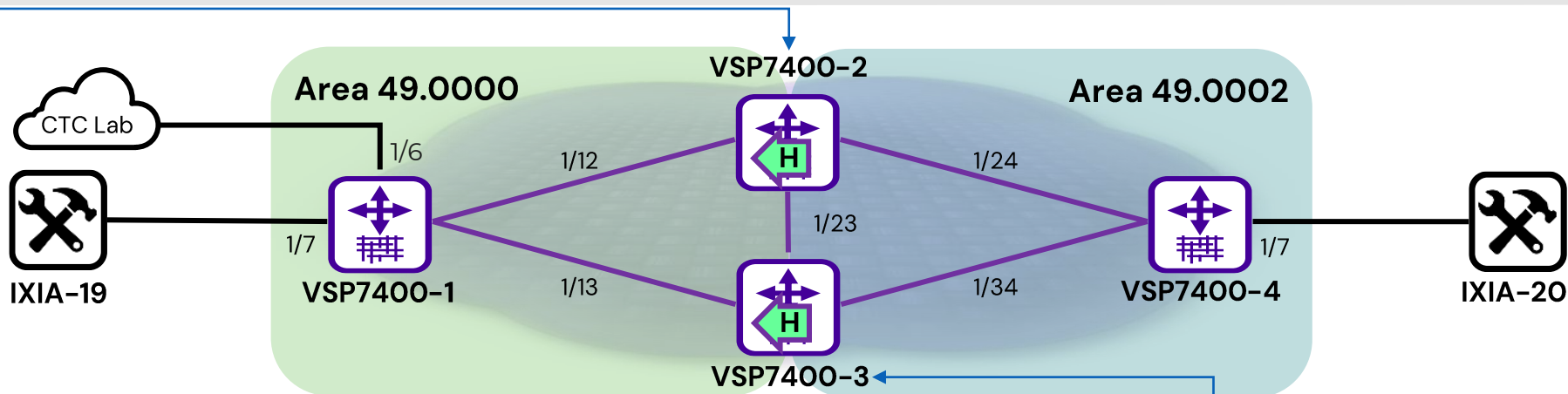
```
Sbox-VSP7400-2:1#[mabn]% show isis|^s+System ID
=====
ISIS General Info
=====
System ID : 02bb.0000.7200
Sbox-VSP7400-2:1#[mabn]% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : e2bb.0000.7200
Multi-Area OperState : non-designated
Sbox-VSP7400-2:1#[mabn]% show isis area-vnode
=====
```

VNODE SYSTEM-ID	VNODE NICK-NAME	VNODE HOST-NAME	REPRESENTED AREA-ADDRESS	REPRESENTED AREA	VNODE USED-IN-AREA
9200.00ff.fff0	9.00.00	Vnode-49.0000	49.0000	HOME	REMOTE
9200.02ff.fff0	9.00.02	Vnode-49.0002	49.0002	REMOTE	HOME

```
Sbox-VSP7400-3:1#% show isis|^s+System ID
=====
ISIS General Info
=====
System ID : 02bb.0000.7300
Sbox-VSP7400-3:1#% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : e2bb.0000.7300
Multi-Area OperState : designated
Sbox-VSP7400-3:1#% show isis area-vnode
=====
```

ISIS SPBM Multi-Area VNode Info					
VNODE SYSTEM-ID	VNODE NICK-NAME	VNODE HOST-NAME	REPRESENTED AREA-ADDRESS	REPRESENTED AREA	VNODE USED-IN-AREA
9200.00ff.fff0	9.00.00	Vnode-49.0000	49.0000	HOME	REMOTE
9200.02ff.fff0	9.00.02	Vnode-49.0002	49.0002	REMOTE	HOME

Inception: status



```
Sbox-VSP7400-2:1#[mabn]% show isis spbm
```

ISIS SPBM Info										
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING	ORIGIN
1	4051-4052	4051	1.00.72	disable	enable	disable	disable	disable	disable	config

ISIS SPBM SMLT Info				
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME
1	primary	02:bb:00:00:72:ff	02bb.0000.7300	1.00.73

```
ISIS Remote SPBM Info
```

SPBM INSTANCE	NICK NAME
1	2.00.72

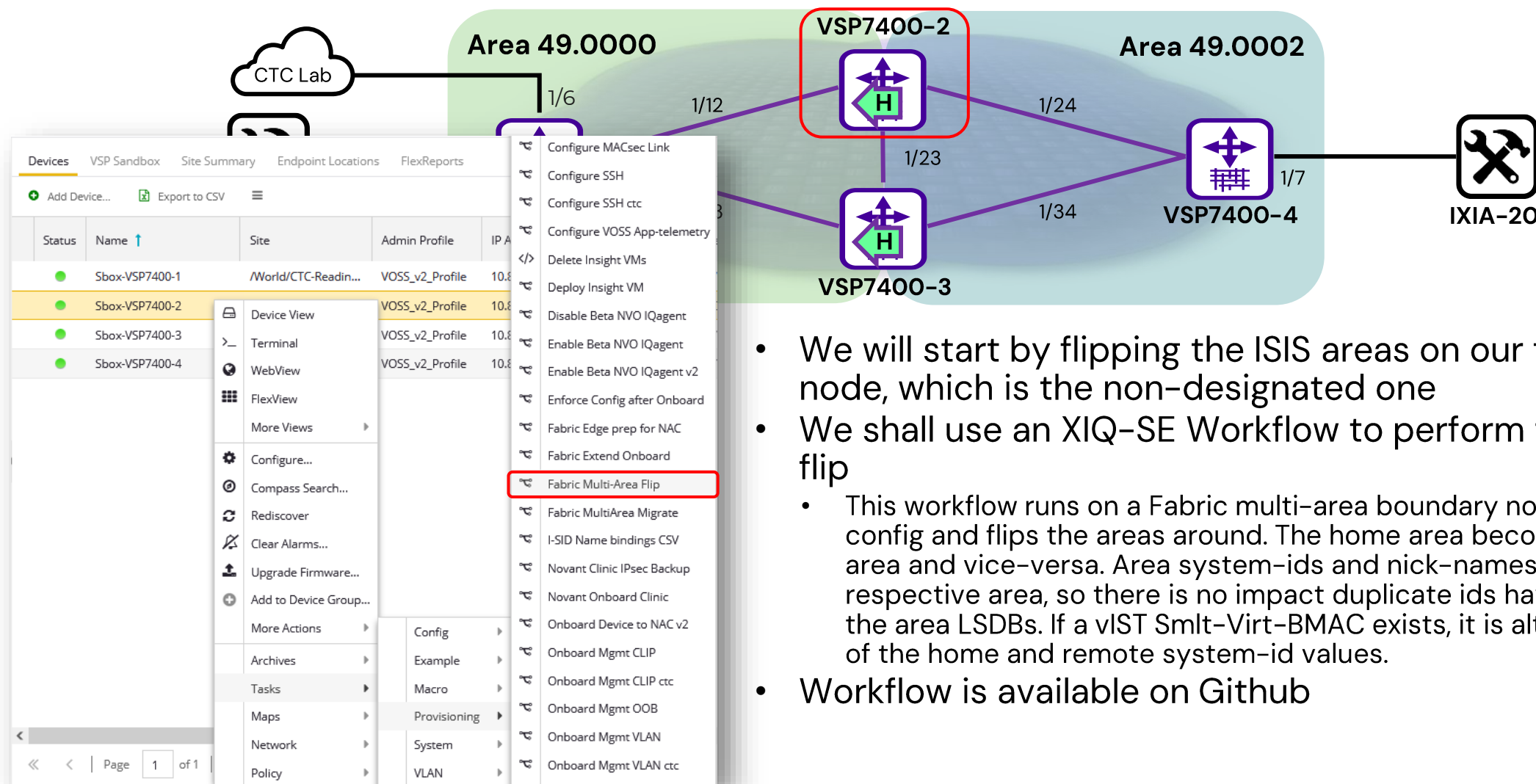
```
Sbox-VSP7400-3:1#% show isis spbm
```

ISIS SPBM Info										
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING	ORIGIN
1	4051-4052	4051	1.00.73	disable	enable	disable	disable	disable	disable	config

ISIS SPBM SMLT Info				
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME
1	secondary	02:bb:00:00:72:ff	02bb.0000.7200	1.00.72

ISIS Remote SPBM Info				
SPBM INSTANCE	NICK NAME			
1	2.00.73			

Action: Flipping areas on first boundary node



- We will start by flipping the ISIS areas on our first boundary node, which is the non-designated one
- We shall use an XIQ-SE Workflow to perform the multi-area flip
 - This workflow runs on a Fabric multi-area boundary node, takes its config and flips the areas around. The home area becomes the remote area and vice-versa. Area system-ids and nick-names are preserved in respective area, so there is no impact duplicate ids having to expire in the area LSDBs. If a vIST Smlt-Virt-BMAC exists, it is altered by an XOR of the home and remote system-id values.
- Workflow is available on Github

Action: Flipping areas on first boundary node

Run Workflow - Fabric Multi-Area Flip

Workflow Inputs

Timeout Properties

Timeout:

4

hr(s)

Custom Inputs

Notes:

This workflow only runs on a Fabric multi-area boundary node. It takes its config and flips the areas around and then reboots the switch on the modified config. The home area becomes the remote area and vice-versa. Area system-ids and nick-names are preserved in respective area, so there is no impact duplicate ids having to expire in the area LSDBs.

If a vIST exists, the Smlt-Virt-BMAC and Smlt-Peer-BMAC are altered by an XOR of the delta bits between home and remote system-id values. Note that if flipping areas of one boundary node first and the second boundary node later, it is important to disable all SMLT interfaces before flipping the areas of the first boundary node; this is because the vIST will not come up while the two boundary nodes have inconsistent areas.

If the switch is managed either via mgmt VLAN or CLIP interface the workflow will check and warn if there is no multi-area redistribution present needed to maintain switch connectivity after restart (note that the same multi-area redistribution must be configured on all boundary nodes, not just the ones having the areas flipped).

Note that direct ISIS adjacencies between boundary nodes for both home and remote areas will not come up while the boundary nodes have inconsistent areas.

The workflow will backup the original config as config.areaflip.bak and the modified config will be stored on /intflash as config.areaflip.cfg.

If you want the workflow only to produce the flipped config and not to proceed with overwriting config.cfg and rebooting the switch, simply enable Sanity and then inspect the above config files.

Sanity and Debug Notes:

Sanity: enable if you do not trust this workflow and wish to first see what it does. In sanity mode configuration changes are not actually made. Debug: enable if you need to report a problem to the script author.

Sanity:

disable

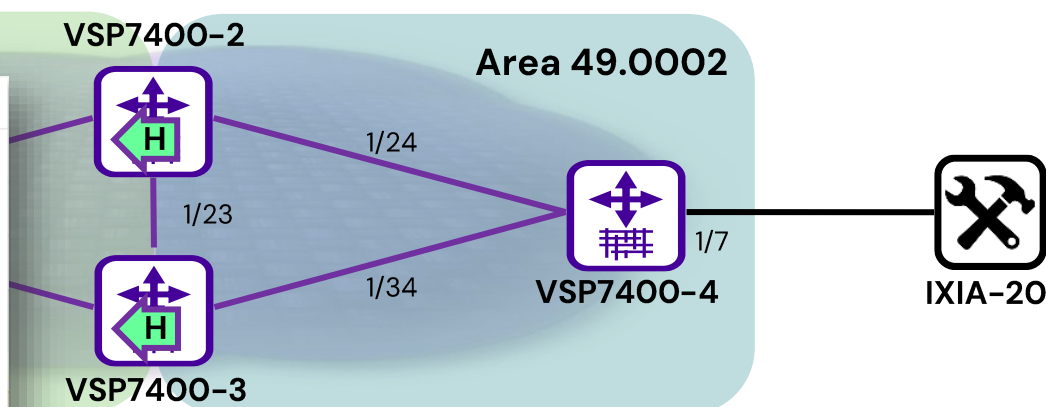
Debug:

enable

Previous

Next

Cancel



- There are no inputs to the workflow.
- The workflow does this:
 1. Performs save config on the boundary node
 2. Retrieves config.cfg
 3. Modifies the config to flip the areas
 4. Makes a backup of config.cfg as config.areaflip.bak
 5. Places modified config on switch as config.areaflip.cfg
 6. Overwrites config.cfg with config.areaflip.cfg
 7. Resets the switch
- To skip the last 2 steps (6,7) enable the Sanity pull down
- NOTE: if the boundary node has a vIST and SMLT connections, it is important to disable all the SMLT links BEFORE running this workflow

Action: Flipping areas on first boundary node

Dashboard
Devices
Discovered
Firmware
Archives
Configuration Templates
Reports
Fabric Multi-Area Flip (9412)

Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	5/13/2026 11:23:08...	Fabric Multi-Area Flip	35	Workflow Designer ...	1	Istevens	5/13/2026 11:23:24...	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOME:49.0002 REMOTE:49.0000 on multi-area boundary node 10.8.4.7	/Work...

Graph View
Table View



Stop Workflow

Show Output

Show Variables

Devices Grid

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	10.8.4.7		5/13/2026 11:...	5/13/2026 11:23:23 AM	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOM...

Show Output

Output - 10.8.4.7

```

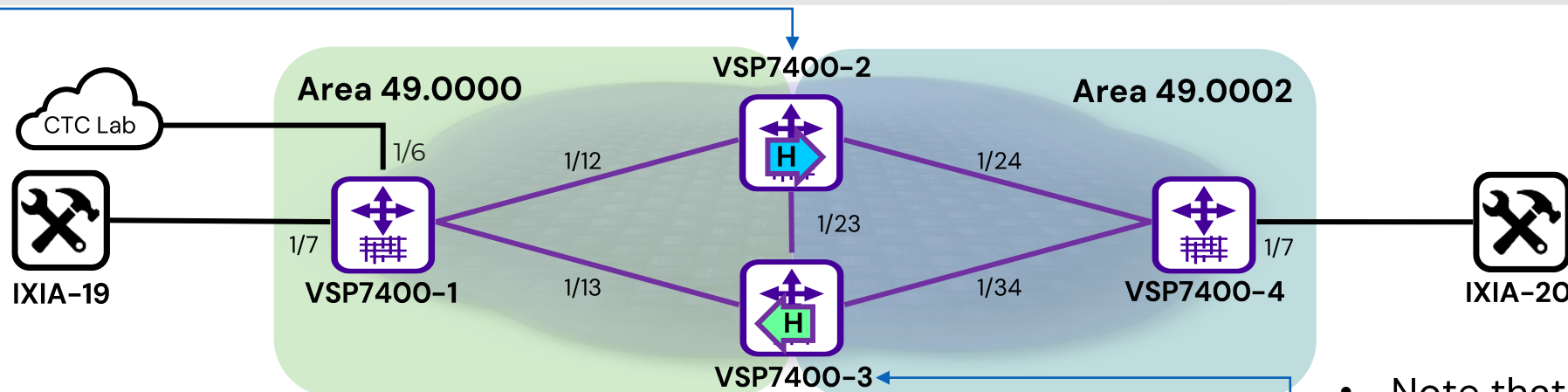
Collected Data:
- HOME: Area = 49.0000
- HOME: Area name = green-area
- HOME: System-id = 02bb.0000.7200
- HOME: Nickname = 1.00.72
- HOME: V-node System-id = 9200.02ff.fff0
- HOME: V-node Nickname = 9.00.02
- HOME: V-node Sys-name = Vnode-49.0002
- HOME: vIST Smlt-Virt-BMAC = 02:bb:00:00:72:ff
- HOME: vIST Smlt-Peer-BMAC = 02bb.0000.7300

- REMOTE: Area = 49.0002
- REMOTE: Area name = blue-area
- REMOTE: System-id = e2bb.0000.7200
- REMOTE: Nickname = 2.00.72
- REMOTE: V-node System-id = 9200.00ff.fff0
- REMOTE: V-node Nickname = 9.00.00
- REMOTE: V-node Sys-name = Vnode-49.0000

- New HOME: vIST Smlt-Virt-BMAC = e2:bb:00:00:72:ff
- New HOME: vIST Smlt-Peer-BMAC = e2bb.0000.7300
- Switch is managed via mgmt 00B1
          
```

- Note that when flipping the HOME & REMOTE areas the workflow will preserve the switch's system-id and nickname in the same areas, so that there is no conflict and no need to wait out 20 minutes for the ISIS LSDB to age out the same IDs in the adjacent areas
- However, if a vIST is present, the Smlt-Virt-BMAC (and Smlt-Peer-BMAC) will need to change. The workflow automatically modifies them in the same way as HOME and REMOTE system-id differ

Migration: After flipping areas on first boundary node



```
Sbox-VSP7400-2:1#[mabn]% show isis area
```

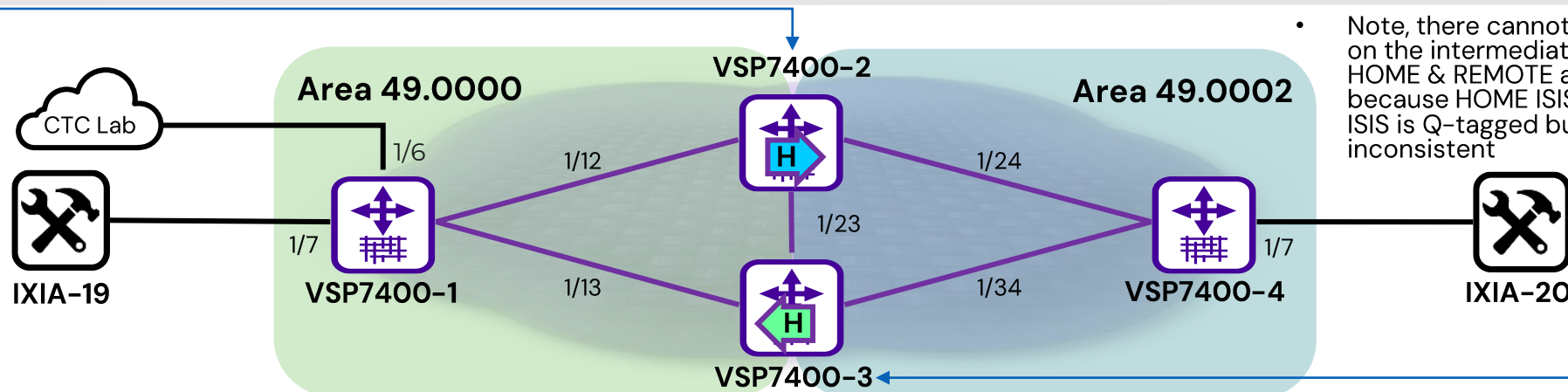
ISIS Area Address			
ADDRESS	ORIGIN	AREA	AREA-NAME
49.0002	config	HOME	blue-area
49.0000	config	REMOTE	green-area

```
Sbox-VSP7400-3:1#% show isis area
```

ISIS Area Address			
ADDRESS	ORIGIN	AREA	AREA-NAME
49.0000	config	HOME	green-area
49.0002	config	REMOTE	blue-area

- Note that this interim situation of having both boundary nodes with inconsistent HOME & REMOTE areas is really an unsupported configuration
- But it can be used to minimize the downtime when migrating to the desired target HOME/REMOTE area configuration

Migration: After flipping areas on first boundary node



- Note, there cannot be any ISIS adjacencies on the intermediate link 1/23 which has both HOME & REMOTE areas configured; this is because HOME ISIS is untagged and REMOTE ISIS is Q-tagged but the area ids are inconsistent

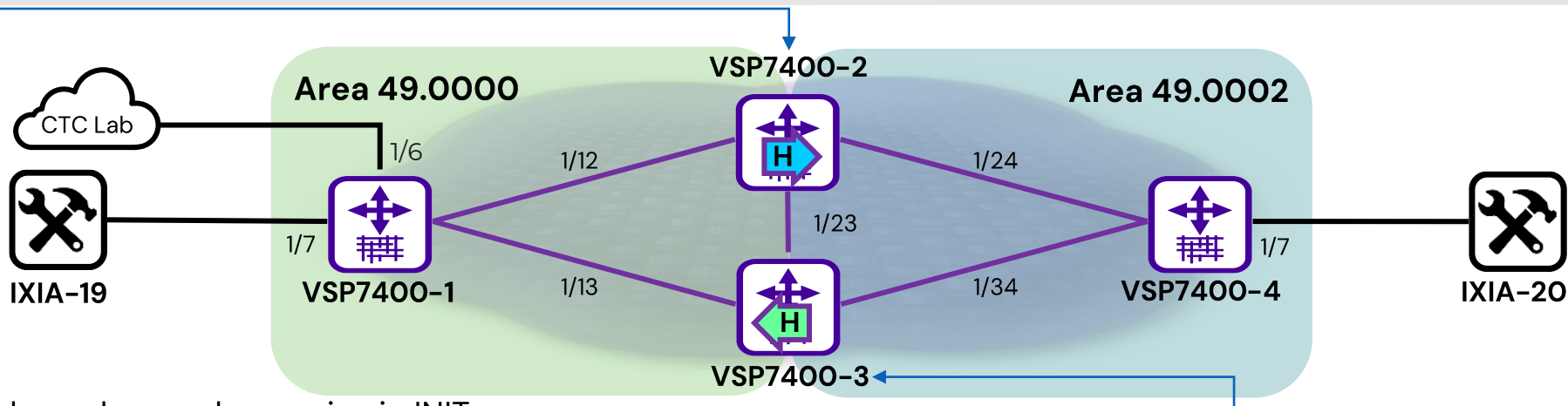
```
Sbox-VSP7400-2:1#[mabn]% show isis adjacencies
```

ISIS Adjacencies										
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA	AREA-NAME
Port1/24	1	UP	00:05:35	127		20 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	HOME	blue-area
Port1/12	1	UP	00:05:35	127		20 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	REMOTE	green-area

```
Sbox-VSP7400-3:1#% show isis adjacencies
```

ISIS Adjacencies										
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA	AREA-NAME
Port1/13	1	UP	00:39:56	127		26 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	HOME	green-area
PortVirtual	1	UP	00:39:43	127		27 9200.02ff.fff0	Vnode-49.0002	ACTIVE	HOME	green-area
Port1/34	1	UP	00:39:53	127		24 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	REMOTE	blue-area
PortVirtual	1	UP	00:39:43	127		27 9200.00ff.fff0	Vnode-49.0000	ACTIVE	REMOTE	blue-area

Migration: After flipping areas on first boundary node

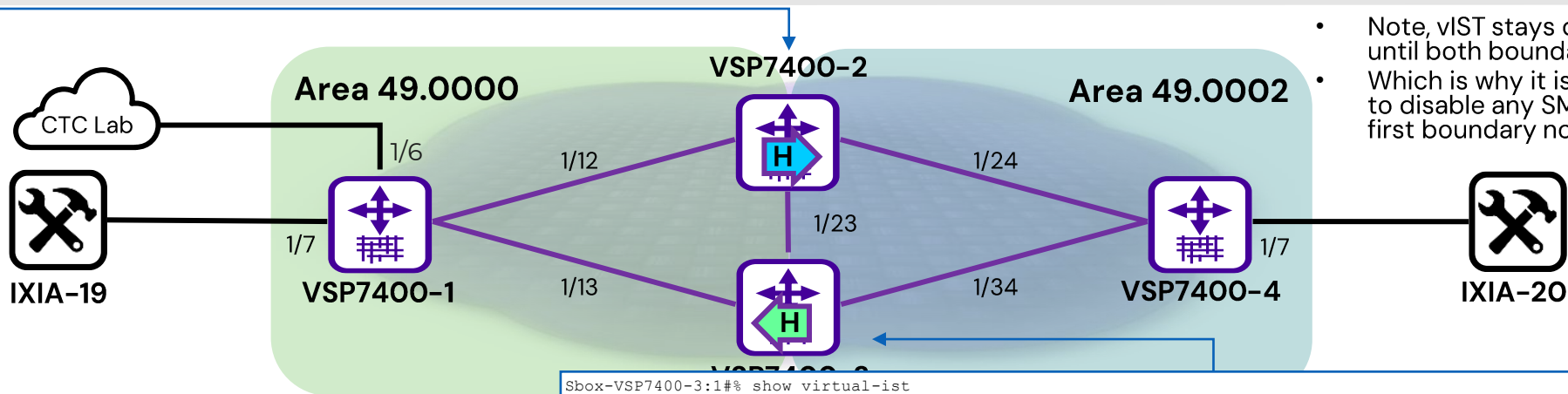


- Note, that the boundary node remains in INIT state because it has inconsistent HOME & REMOTE areas compared to the designated boundary node which we have not yet flipped

```
Sbox-VSP7400-2:1#[mabn]% show isis|^s+System ID
=====
ISIS General Info
=====
System ID : e2bb.0000.7200
Sbox-VSP7400-2:1#[mabn]% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : 02bb.0000.7200
Multi-Area OperState : init
Sbox-VSP7400-2:1#[mabn]% show isis area-vnode
=====
ISIS SPBM Multi-Area VNode Info
=====
VNODE      VNODE      VNODE
SYSTEM-ID  NICK-NAME  HOST-NAME
-----
9200.02ff.fff0  9.00.02    Vnode-49.0000
9200.00ff.fff0  9.00.00    Vnode-49.0002
```

```
Sbox-VSP7400-3:1#% show isis|^s+System ID
=====
ISIS General Info
=====
System ID : 02bb.0000.7300
Sbox-VSP7400-3:1#% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : e2bb.0000.7300
Multi-Area OperState : designated
Sbox-VSP7400-3:1#% show isis area-vnode
=====
ISIS SPBM Multi-Area VNode Info
=====
VNODE      VNODE      VNODE      REPRESENTED  REPRESENTED  VNODE
SYSTEM-ID  NICK-NAME  HOST-NAME  AREA-ADDRESS  AREA        USED-IN-AREA
-----
9200.00ff.fff0  9.00.00    Vnode-49.0000  49.0000      HOME        REMOTE
9200.02ff.fff0  9.00.02    Vnode-49.0002  49.0002      REMOTE      HOME
```

Migration: After flipping areas on first boundary node



- Note, vIST stays down and will not come up until both boundary nodes are flipped
- Which is why it is important, to avoid loops, to disable any SMLT ports before flipping the first boundary node areas

Sbox-VSP7400-2:1#[mabn]% show virtual-ist

IST Info			
PEER-IP ADDRESS	VLAN ID	ENABLE IST	IST STATUS
192.168.255.2	4000	true	down
NEGOTIATED DIALECT	IST STATE	MASTER/SLAVE	
NONE	Down	Slave	

Sbox-VSP7400-2:1#[mabn]% show isis spbm

ISIS SPBM Info									
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING
1	4051-4052	4051	2.00.72	disable	enable	disable	disable	disable	config
ISIS SPBM SMLT Info									
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME	SMLT-PEER-HOST-NAME				
1	primary	e2:bb:00:00:72:ff	e2bb.0000.7300	2.00.73	Sbox-VSP7400-3				

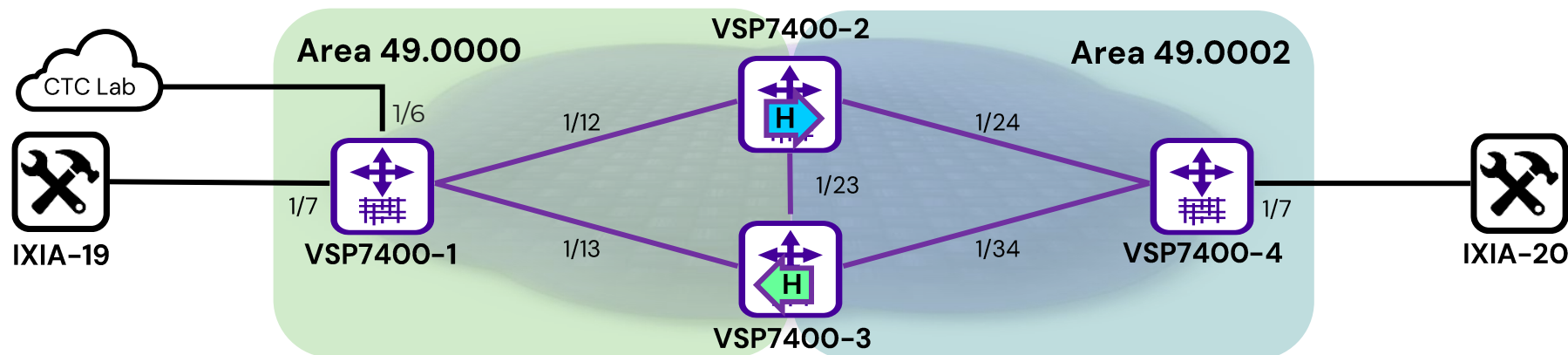
Sbox-VSP7400-3:1#% show virtual-ist

IST Info			
PEER-IP ADDRESS	VLAN ID	ENABLE IST	IST STATUS
192.168.255.1	4000	true	down
NEGOTIATED DIALECT	IST STATE	MASTER/SLAVE	
NONE	Down	Master	

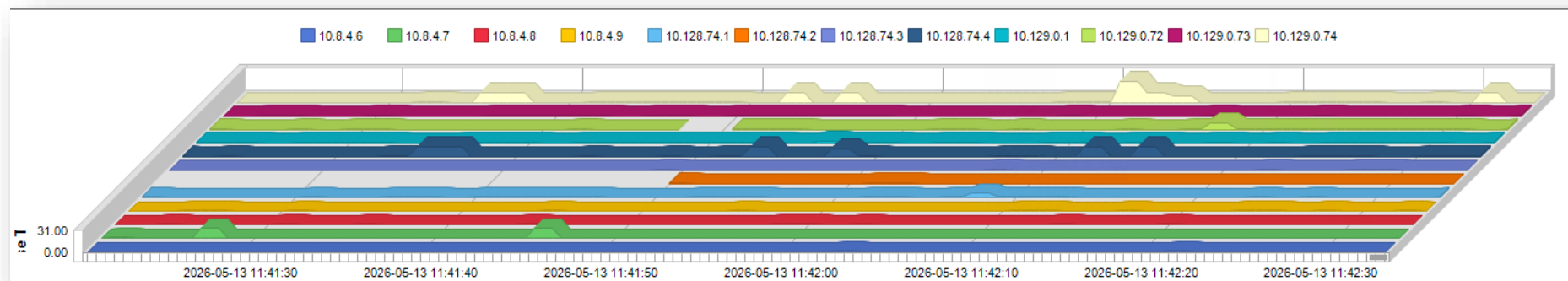
Sbox-VSP7400-3:1#% show isis spbm

ISIS SPBM Info										
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING	ORIGIN
1	4051-4052	4051	1.00.73	disable	enable	disable	disable	disable	disable	config
ISIS SPBM SMLT Info										
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME	SMLT-PEER-HOST-NAME					
1	secondary	02:bb:00:00:72:ff	02bb.0000.7200	1.00.72	Sbox-VSP7400-2					

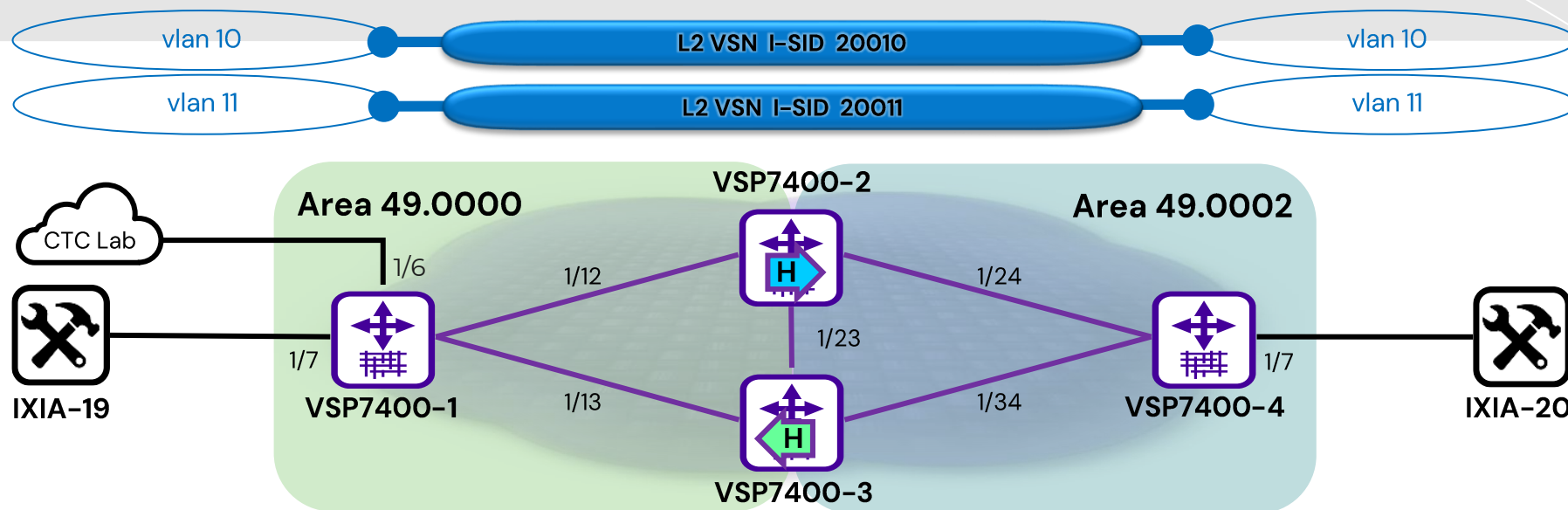
Migration: After flipping areas on first boundary node



- Boundary node VSP7400-2 becomes reachable again on all mgmt interfaces: OOB, VLAN, CLIP
- For the mgmt VLAN & CLIP, the other designated boundary node VSP7400-3 is ensuring connectivity for them



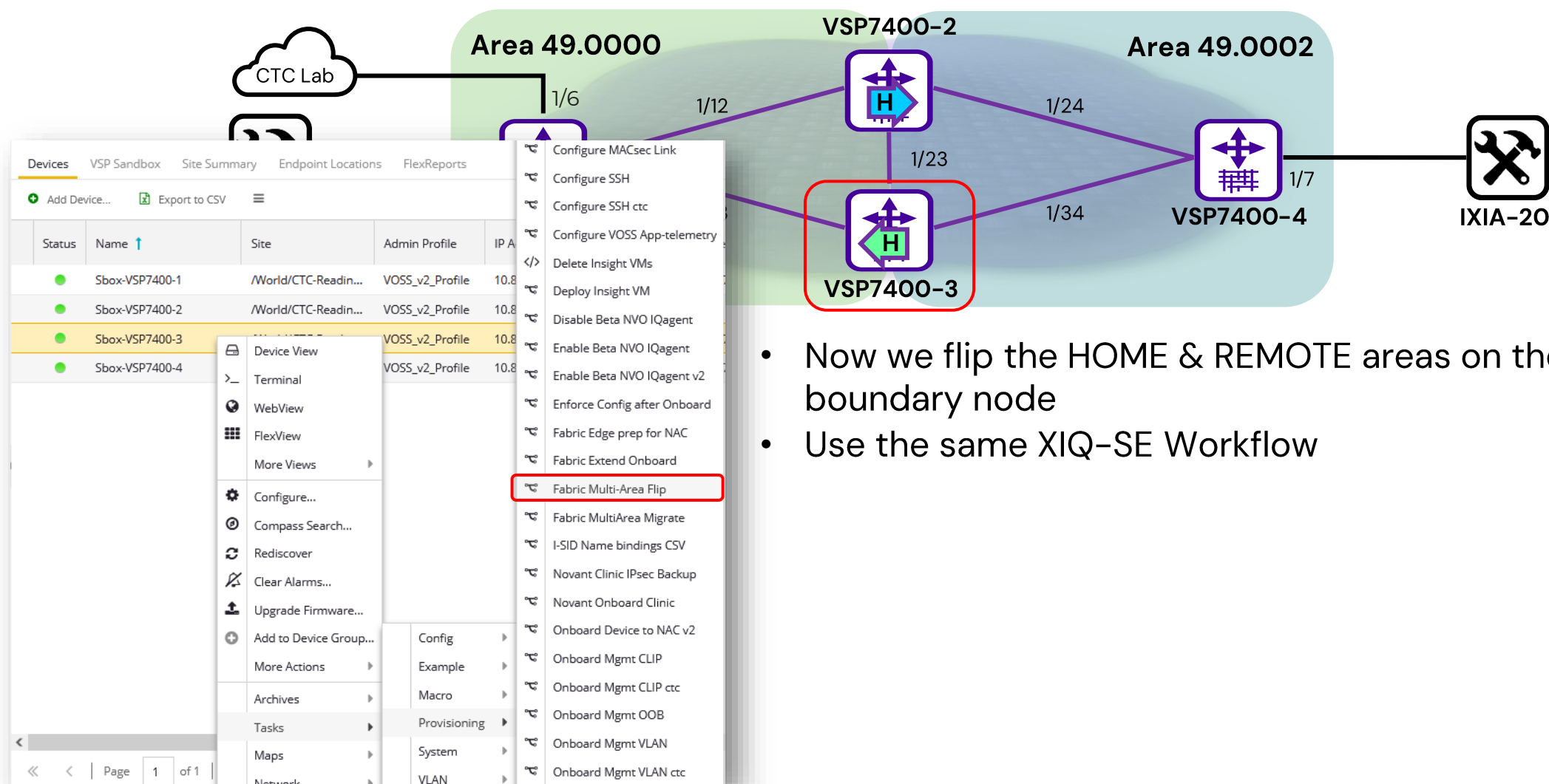
Migration: After flipping areas on first boundary node



PGID	Total # Frames Rec.	Latency Avg (µs)	Byte Count	Frame Rate (/sec)	In Order Packets	Reordered Packets	Duplicate Packets	Late Packets	Lost Packets
Chassis: 10.8.8.2, Card: 01, Port: 019									
1	208,471	10.136	21,680,984	1,000	208,471	0	0	0	0
Chassis: 10.8.8.2, Card: 01, Port: 020									
0	208,351	10.157	21,668,504	1,000	208,351	0	0	0	121

- The reboot of non-designated boundary node VSP7400-2 and it restarting with HOME & REMOTE areas flipped and into INIT state
- This caused a sub-second traffic interruption for those L2VSNs which were hashed over that boundary node

Action: Flipping areas on second boundary node



- Now we flip the HOME & REMOTE areas on the second boundary node
- Use the same XIQ-SE Workflow

Action: Flipping areas on second boundary node

Dashboard
Devices
Discovered
Firmware
Archives
Configuration Templates
Reports
Fabric Multi-Area Flip (9413)
Fabric Multi-Area Flip (9414) ✕

Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message
✓	5/13/2026 12:12:01...	Fabric Multi-Area Flip	35	Workflow Designer ...	1	Istevens	5/13/2026 12:12:16...	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOME:49.0002 REMOTE:49.0000 on multi-area boundary node 10.8.4.8

Graph View
Table View



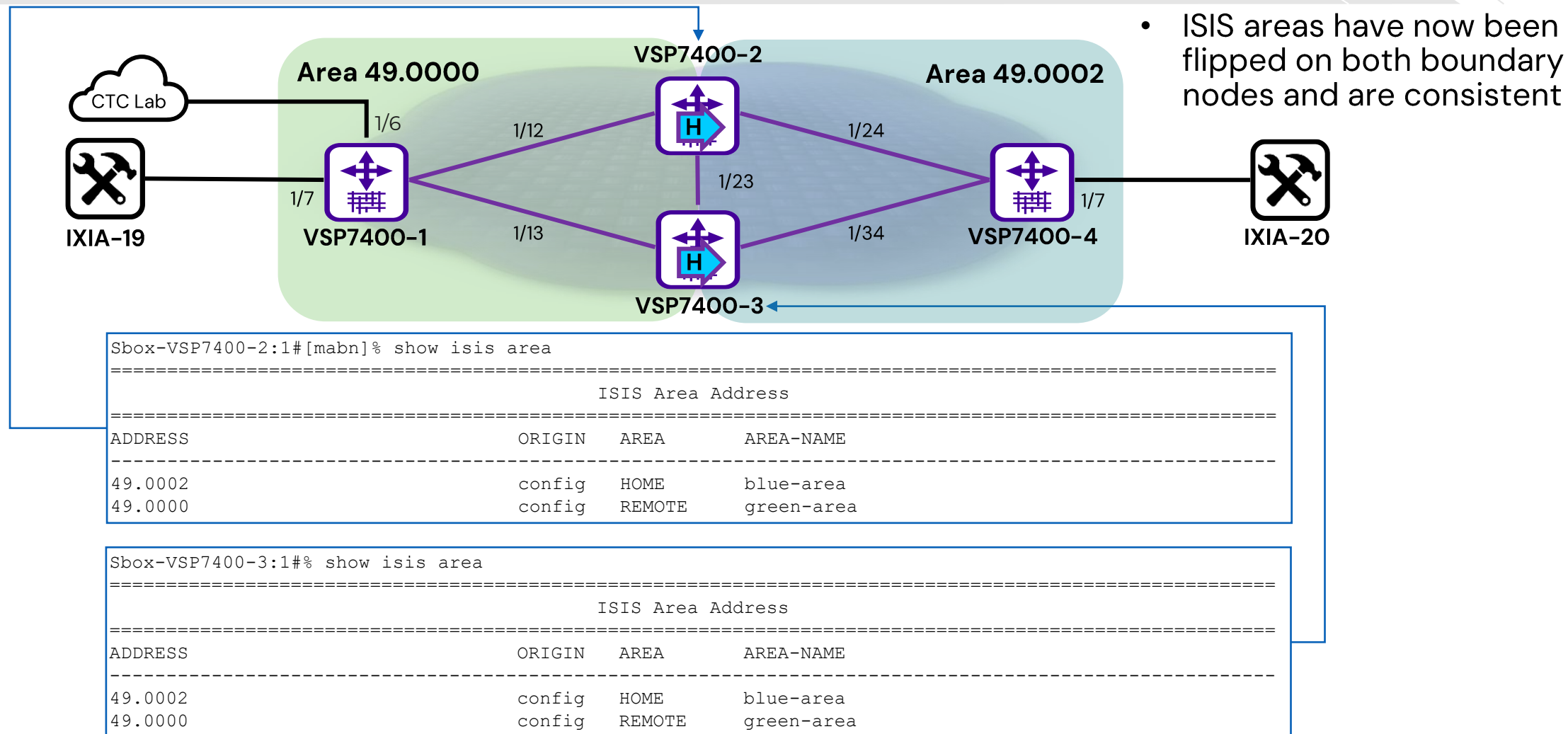
 Stop Workflow
 Show Output
 Show Variables

Devices Grid

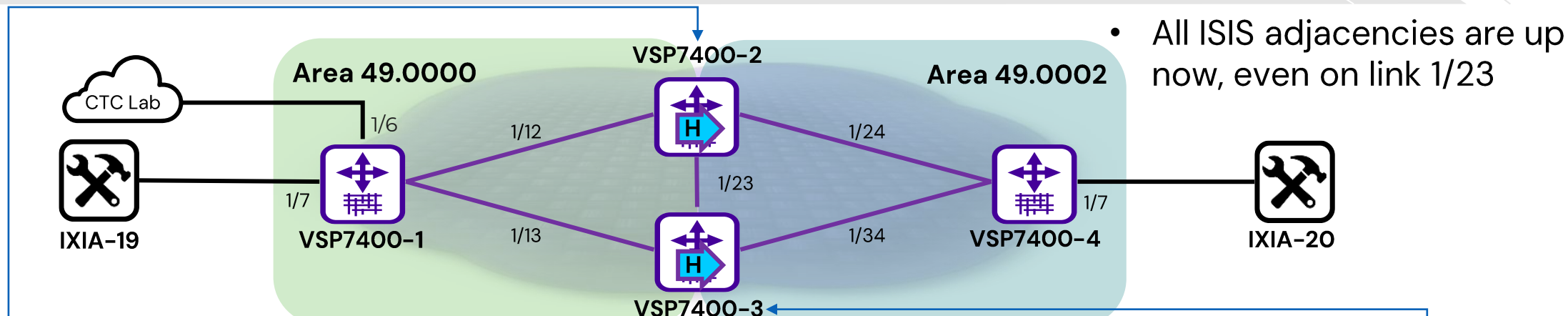
Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	10.8.4.8		5/13/2026 12:...	5/13/2026 12:...	Flipped areas from HOME:49.0000 REMOTE:49.0

- As soon as the workflow has completed, and the second boundary node is rebooted, if there is a vIST between the boundary nodes, now is the time to go and re-enable all SMLT links on the first boundary node which we already flipped

Target: After flipping areas on second boundary node



Target: After flipping areas on second boundary node



- All ISIS adjacencies are up now, even on link 1/23

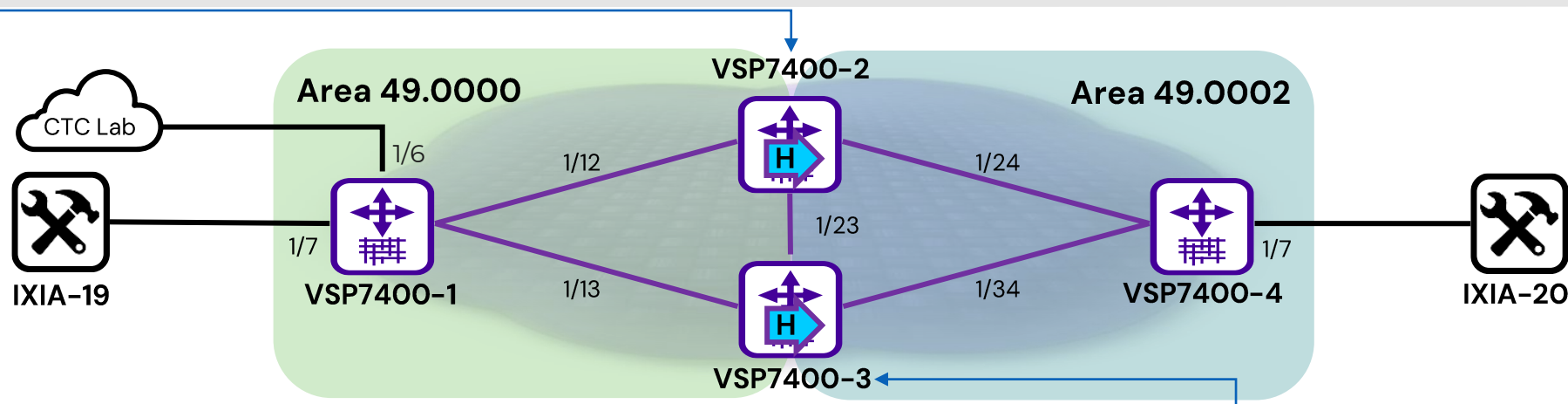
```
Sbox-VSP7400-2:1#[mabn]% show isis adjacencies
```

ISIS Adjacencies									
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA
Port1/23	1	UP	00:05:37	127		20 e2bb.0000.7300	Sbox-VSP7400-3	ACTIVE	HOME
Port1/24	1	UP	00:39:51	127		24 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	HOME
PortVirtual	1	UP	00:07:59	127		27 9200.00ff.fff0	Vnode-49.0002	ACTIVE	HOME
Port1/12	1	UP	00:39:51	127		19 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	REMOTE
Port1/23	1	UP	00:05:37	127		23 02bb.0000.7300	Sbox-VSP7400-3	ACTIVE	REMOTE
PortVirtual	1	UP	00:07:59	127		27 9200.02ff.fff0	Vnode-49.0000	ACTIVE	REMOTE

```
Sbox-VSP7400-3:1#% show isis adjacencies
```

ISIS Adjacencies									
INTERFACE	L	STATE	UPTIME	PRI	HOLDTIME	SYSID	HOST-NAME	STATUS	AREA
Port1/23	1	UP	00:05:37	127		26 e2bb.0000.7200	Sbox-VSP7400-2	ACTIVE	HOME
Port1/34	1	UP	00:05:37	127		24 02bb.0000.7400	Sbox-VSP7400-4	ACTIVE	HOME
PortVirtual	1	UP	00:05:25	127		27 9200.00ff.fff0	Vnode-49.0002	ACTIVE	HOME
Port1/13	1	UP	00:05:37	127		23 02bb.0000.7100	Sbox-VSP7400-1	ACTIVE	REMOTE
Port1/23	1	UP	00:05:37	127		20 02bb.0000.7200	Sbox-VSP7400-2	ACTIVE	REMOTE
PortVirtual	1	UP	00:05:25	127		27 9200.02ff.fff0	Vnode-49.0000	ACTIVE	REMOTE

Target: After flipping areas on second boundary node



- Both boundary nodes are now active and operational, one in designated state the other in non-designated state

```
Sbox-VSP7400-2:1#[mabn]% show isis|^s+System ID
=====
ISIS General Info
=====
System ID : e2bb.0000.7200
Sbox-VSP7400-2:1#[mabn]% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : 02bb.0000.7200
Multi-Area OperState : non-designated
Sbox-VSP7400-2:1#[mabn]% show isis area-vnode
=====
```

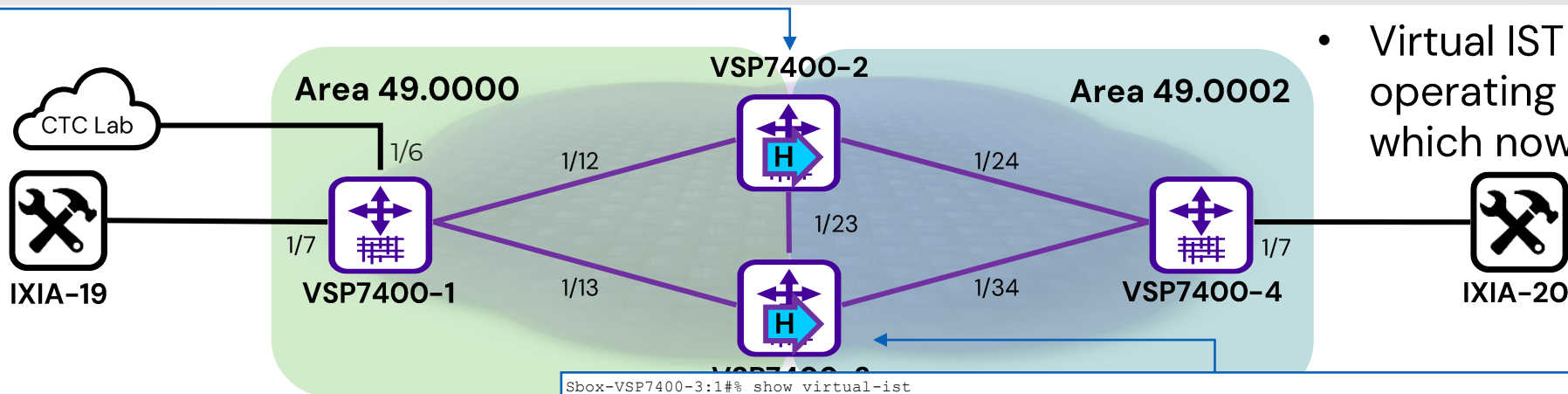
VNODE SYSTEM-ID	VNODE NICK-NAME	VNODE HOST-NAME	REPRESENTED AREA-ADDRESS	REPRESENTED AREA	VNODE USED-IN-AREA
9200.02ff.fff0	9.00.02	Vnode-49.0000	49.0002	HOME	REMOTE
9200.00ff.fff0	9.00.00	Vnode-49.0002	49.0000	REMOTE	HOME

```
Sbox-VSP7400-3:1#% show isis|^s+System ID
```

```
=====
ISIS General Info
=====
System ID : e2bb.0000.7300
Sbox-VSP7400-3:1#% show isis remote|^s+System ID,OperState
=====
ISIS Remote Area Info
=====
System ID : 02bb.0000.7300
Multi-Area OperState : designated
Sbox-VSP7400-3:1#% show isis area-vnode
=====
```

ISIS SPBM Multi-Area VNode Info					
VNODE SYSTEM-ID	VNODE NICK-NAME	VNODE HOST-NAME	REPRESENTED AREA-ADDRESS	REPRESENTED AREA	VNODE USED-IN-AREA
9200.02ff.fff0	9.00.02	Vnode-49.0000	49.0002	HOME	REMOTE
9200.00ff.fff0	9.00.00	Vnode-49.0002	49.0000	REMOTE	HOME

Target: After flipping areas on second boundary node



- Virtual IST is up and now operating in area 49.0002 which now the HOME area

Sbox-VSP7400-2:1#[mabn]% show virtual-ist

IST Info			
PEER-IP ADDRESS	VLAN ID	ENABLE IST	IST STATUS
192.168.255.2	4000	true	up
NEGOTIATED DIALECT	IST STATE	MASTER/SLAVE	
v6.0	Up	Slave	

Sbox-VSP7400-2:1#[mabn]% show isis spbm

ISIS SPBM Info									
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING
1	4051-4052	4051	2.00.72	disable	enable	disable	disable	disable	config
ISIS SPBM SMLT Info									
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME	SMLT-PEER-HOST-NAME				
1	primary	e2:bb:00:00:72:ff	e2bb.0000.7300	2.00.73	Sbox-VSP7400-3				

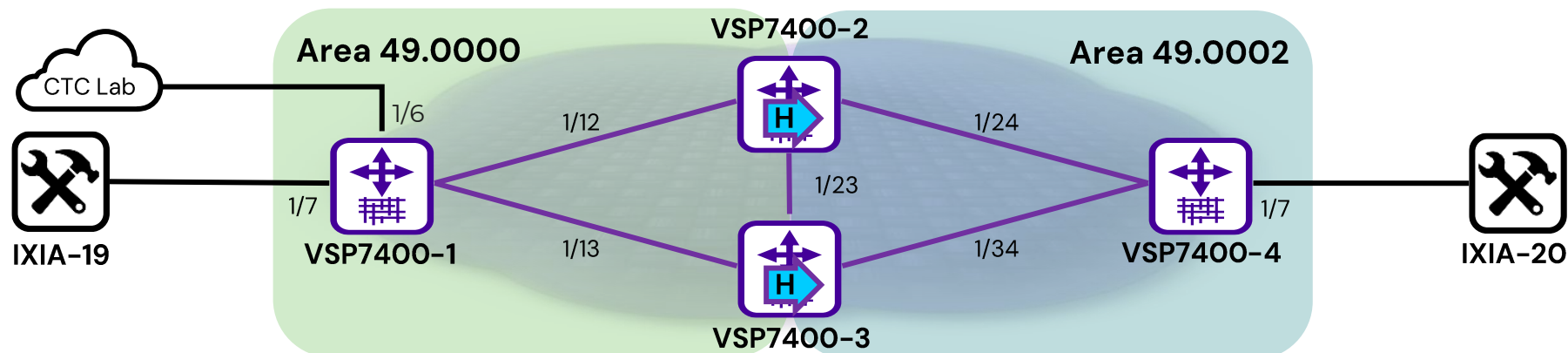
Sbox-VSP7400-3:1#% show virtual-ist

IST Info			
PEER-IP ADDRESS	VLAN ID	ENABLE IST	IST STATUS
192.168.255.1	4000	true	up
NEGOTIATED DIALECT	IST STATE	MASTER/SLAVE	
v6.0	Up	Master	

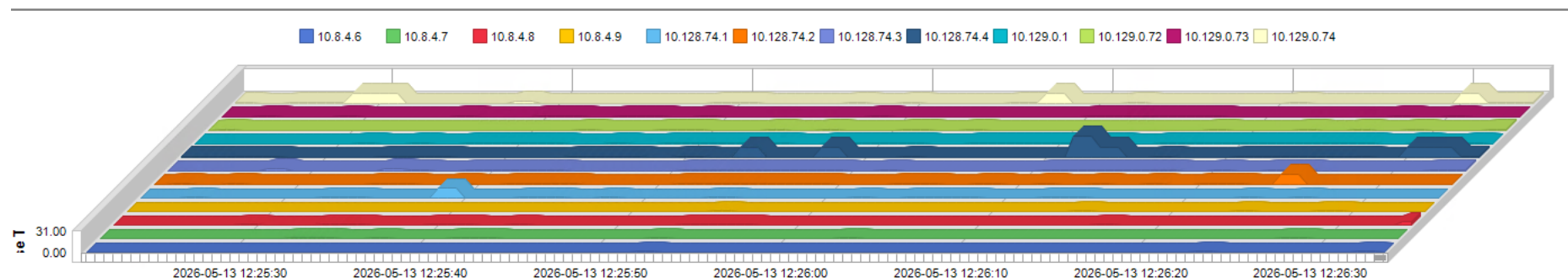
Sbox-VSP7400-3:1#% show isis spbm

ISIS SPBM Info										
SPBM INSTANCE	B-VID	PRIMARY VLAN	NICK NAME	LSDB TRAP	IP	IPV6	MULTICAST	SPB-PIM-GW	STP-MULTI HOMING	ORIGIN
1	4051-4052	4051	2.00.73	disable	enable	disable	disable	disable	disable	config
ISIS SPBM SMLT Info										
SPBM INSTANCE	SMLT-SPLIT-BEB	SMLT-VIRTUAL-BMAC	SMLT-PEER-SYSTEM-ID	SMLT-PEER-NICK-NAME	SMLT-PEER-HOST-NAME					
1	secondary	e2:bb:00:00:72:ff	e2bb.0000.7200	2.00.72	Sbox-VSP7400-2					

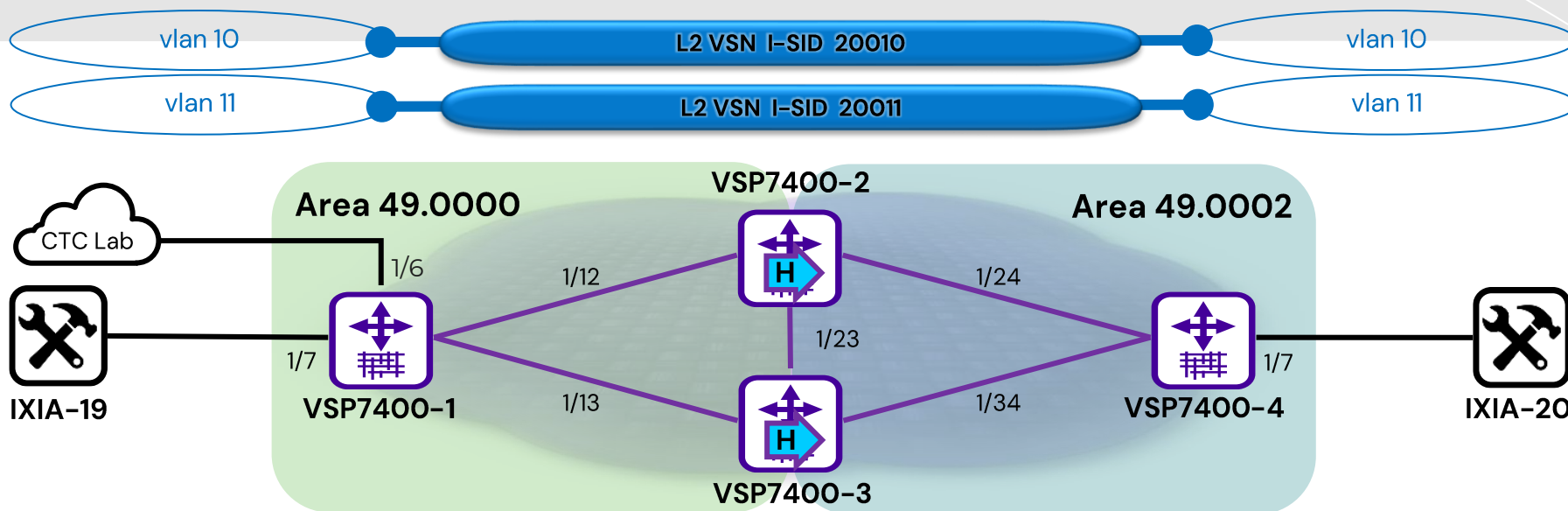
Target: After flipping areas on second boundary node



- Both boundary nodes become reachable again on all mgmt interfaces: OOB, VLAN, CLIP



Target: After flipping areas on second boundary node



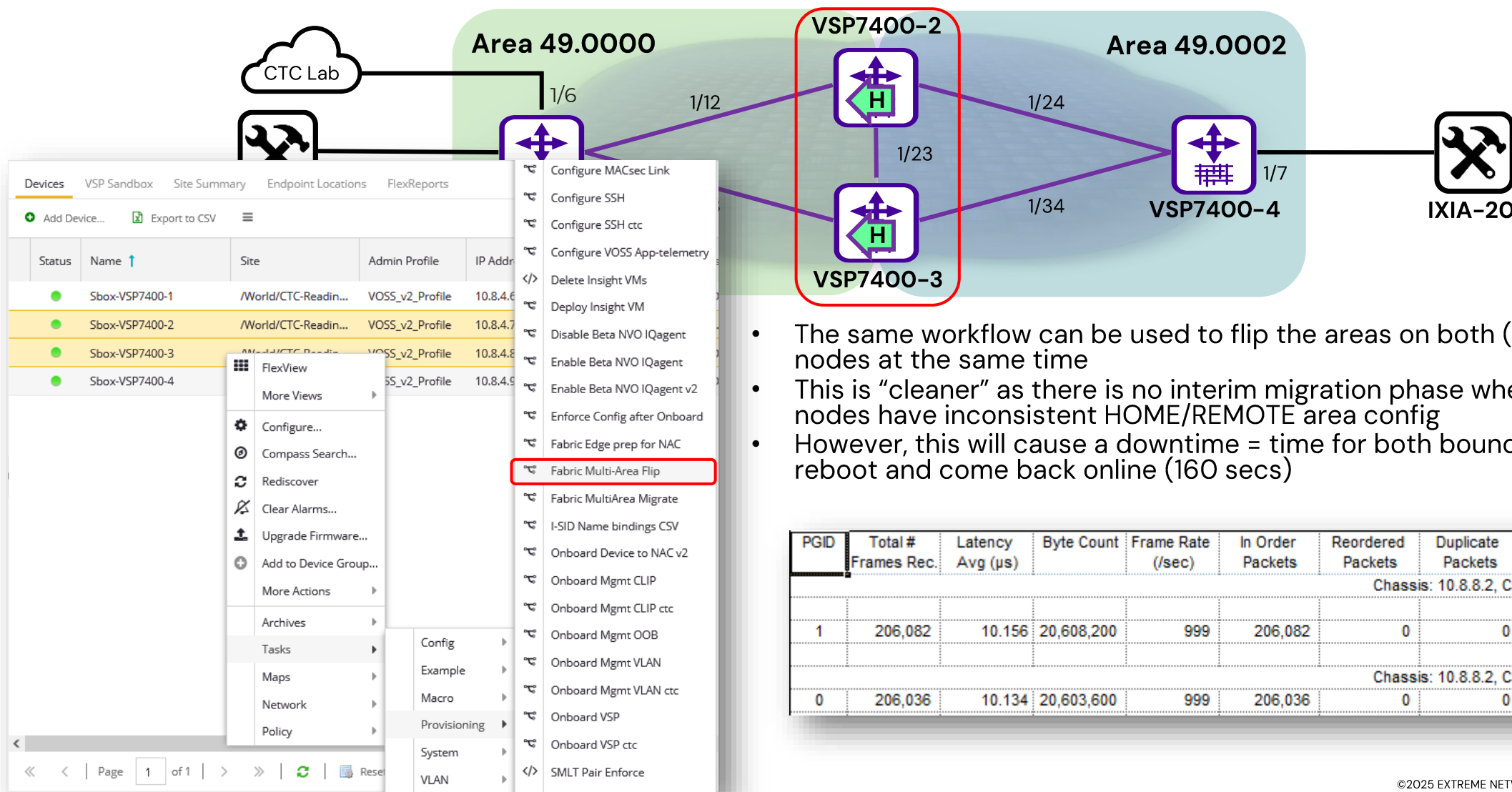
PGID	Total # Frames Rec.	Latency Avg (µs)	Byte Count	Frame Rate (/sec)	In Order Packets	Reordered Packets	Duplicate Packets	Late Packets	Lost Packets
Chassis: 10.8.8.2, Card: 01, Port: 019									
1	248,767	10.142	25,871,768	999	248,767	0	0	0	2,969
Chassis: 10.8.8.2, Card: 01, Port: 020									
0	248,708	10.153	25,865,632	999	248,708	0	0	0	3,029

- The reboot of the VSP7400-3 designated boundary node + the first VSP7400-2 boundary node transitioning from INIT → Designated state is just under 3 seconds
- And when the VSP7400-3 boundary comes back online there is a sub-second interruption for it to become Designated once again
- All in all, the migration only had a 3 seconds hit for the L2VSN traffic

Variants

- The same steps will also work with 4 boundary nodes between the same two areas. In this case no Virtual-IST will exist on those boundary nodes.
- These tests find that the best approach, to minimize downtime, is to flip the areas on all the non-designated boundary nodes first and then flip the areas on the Designated boundary node last
- Flipping the Designated boundary node first is possible but this will immediately have an impact for traffic, rather than having an impact later if the Designated boundary was flipped last. The impact recorded is greater: 300-400 milli-seconds on reboot and then 5 secs on boot up

Variant: Flipping areas on both (all) boundary nodes at same time



- The same workflow can be used to flip the areas on both (all) boundary nodes at the same time
- This is “cleaner” as there is no interim migration phase where the boundary nodes have inconsistent HOME/REMOTE area config
- However, this will cause a downtime = time for both boundary nodes to reboot and come back online (160 secs)

PGID	Total # Frames Rec.	Latency Avg (µs)	Byte Count	Frame Rate (/sec)	In Order Packets	Reordered Packets	Duplicate Packets	Late Packets	Lost Packets
Chassis: 10.8.8.2, Card: 01, Port: 019									
1	206,082	10.156	20,608,200	999	206,082	0	0	0	158,296
Chassis: 10.8.8.2, Card: 01, Port: 020									
0	206,036	10.134	20,603,600	999	206,036	0	0	0	158,342

Variant: Flipping areas on both (all) boundary nodes at same time

Dashboard
Devices
Discovered
Firmware
Archives
Configuration Templates
Reports
Fabric Multi-Area Flip (9417)

Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message
✓	5/13/2026 1:33:57 ...	Fabric Multi-Area Flip	37	Workflow Designer ...	2	Istevens	5/13/2026 1:34:14 ...	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOME:49.0002 REMOTE:49.0000 on multi-area boundary nodes 10.8.4.7, 10.8.4.8

Graph View
Table View



Stop Workflow

Show Output

Show Variables

Devices Grid

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	10.8.4.7		5/13/2026 1:3...	5/13/2026 1:3...	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOME:49.0002 R
SUCCESS	10.8.4.8		5/13/2026 1:3...	5/13/2026 1:3...	Flipped areas from HOME:49.0000 REMOTE:49.0002 to HOME:49.0002 R

Output - 10.8.4.7

```

Collected Data:
- HOME: Area = 49.0000
- HOME: Area name = green-area
- HOME: System-id = 02bb.0000.7200
- HOME: Nickname = 1.00.72
- HOME: V-node System-id = 9200.02ff.fff0
- HOME: V-node Nickname = 9.00.02
- HOME: V-node Sys-name = Vnode-49.0002
- HOME: vIST Smlt-Virt-BMAC = 02:bb:00:00:72:ff
- HOME: vIST Smlt-Peer-BMAC = 02bb.0000.7300

- REMOTE: Area = 49.0002
- REMOTE: Area name = blue-area
- REMOTE: System-id = e2bb.0000.7200
- REMOTE: Nickname = 2.00.72
- REMOTE: V-node System-id = 9200.00ff.fff0
- REMOTE: V-node Nickname = 9.00.00
- REMOTE: V-node Sys-name = Vnode-49.0000

- New HOME: vIST Smlt-Virt-BMAC = e2:bb:00:00:72:ff
- New HOME: vIST Smlt-Peer-BMAC = e2bb.0000.7300
- Switch is managed via mgmt OOB1

Sbox-VSP7400-2:1#save config

```

Output - 10.8.4.8

```

Collected Data:
- HOME: Area = 49.0000
- HOME: Area name = green-area
- HOME: System-id = 02bb.0000.7300
- HOME: Nickname = 1.00.73
- HOME: V-node System-id = 9200.02ff.fff0
- HOME: V-node Nickname = 9.00.02
- HOME: V-node Sys-name = Vnode-49.0002
- HOME: vIST Smlt-Virt-BMAC = 02:bb:00:00:72:ff
- HOME: vIST Smlt-Peer-BMAC = 02bb.0000.7200

- REMOTE: Area = 49.0002
- REMOTE: Area name = blue-area
- REMOTE: System-id = e2bb.0000.7300
- REMOTE: Nickname = 2.00.73
- REMOTE: V-node System-id = 9200.00ff.fff0
- REMOTE: V-node Nickname = 9.00.00
- REMOTE: V-node Sys-name = Vnode-49.0000

- New HOME: vIST Smlt-Virt-BMAC = e2:bb:00:00:72:ff
- New HOME: vIST Smlt-Peer-BMAC = e2bb.0000.7200
- Switch is managed via mgmt OOB1

```



Extreme[®]
networks